



High-resolution maize yield mapping across Africa using earth observation and machine learning, deep learning, and foundation model

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ABSTRACT

Africa's food security is increasingly threatened by climate change and population growth, creating an urgent need for high-resolution crop yield maps to support precision agriculture and climate adaptation; however, much of the continent remains poorly monitored due to limited ground observations. This study introduces a novel 250-m spatial resolution maize yield prediction framework for 42 African countries. Our methodology utilizes Net Primary Production (NPP) data to spatially disaggregate national FAO yield statistics, generating fine-scale training data for supervised learning. A comprehensive feature set of 296 variables was constructed by integrating multi-source Earth observation, climate, and soil data. We evaluated multiple benchmark models for tabular data, including XGBoost, LightGBM, a Hybrid Deep Neural Network (HDNN), and, for the first time in this context, a tabular foundation model, the Tabular Prior data Fitted Network (TabPFN). Using an expanding-window temporal cross-validation strategy, XGBoost achieved the highest temporal R^2 (0.78), while TabPFN demonstrated superior spatial generalization and the lowest mean absolute percentage error (MAPE $\approx$ 25 %). Causal inference and ablation analyses highlighted the predictive importance of vegetation indices (e.g., NDVI, NDWI), drought metrics, and soil properties. Model outputs closely matched national yields ($R^2 > 0.75$; MAPE $\approx$ 26–28 %). This study provides a scalable framework for yield monitoring in data-scarce regions and represents the first successful application of tabular foundation models in continental-scale agricultural prediction.

1. Introduction

Africa has shown significant progress in improving cereal crop yields over the past decades, primarily driven by increased use of fertilizers, high-yielding seeds, and technological advancements. Average yields have doubled from about 1.1 t ha⁻¹ in the 1960s to 2.2 t ha⁻¹ in the

1990s, further increasing to 4 t ha⁻¹ in 2020 (FAOSTAT, 2021; Le Roux and Cilliers, 2025). Though the increase has been substantial, Africa's productivity is still far behind the global average. Although the general trend in cereal crop yields is upward, there is substantial interannual and spatial variability and this variability is largely explained by fluctuations in climate (Ray et al., 2015; Lesk et al., 2022), differences in soil quality,

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and variations in management practices (Tilman et al., 2011; Mueller et al., 2012; Fan et al., 2011). Countries like Egypt and South Africa achieve cereal yields more than 5 t ha⁻¹, while countries such as Namibia, Niger, and Mozambique have yields below 1 t ha⁻¹, primarily due to limited amount of irrigation, agricultural inputs, and access to technology (FAO, 2023).

Addressing these disparities calls for a detailed understanding of spatial yield dynamics across diverse agro-ecological zones, and therefore, High-resolution crop yield maps are essential for assessing food security risks, identifying untapped crop production potential, and analyzing the localized impacts of extreme weather events and climate change (Grassini et al., 2024). However, in the absence of comprehensive ground-based measurements, such crop yield maps are increasingly generated through the integration of dynamic crop models, remote sensing, and machine learning (ML) techniques. Beyond traditional statistical and process-based modelling approaches, ML and deep learning (DL) methods have emerged as powerful tools for crop yield prediction, enabling the fusion of satellite imagery, soil information, and climate records to predict yields at unprecedented spatial resolutions (Leng and Hall, 2020; Leroux et al., 2019; Adisa et al., 2019; Tamás et al., 2023; Asamoah et al., 2024; Kheir et al., 2024).

Studies across Africa demonstrate that tree-based ensemble ML models such as Random Forest and Gradient Boosting excel at capturing nonlinear relationships between environmental variables and crop yields, achieving R² values of up to 0.63 in countries like Kenya and Zambia (Kaneko et al., 2019) and root mean square error (RMSE) values as low as 1.76–3.94 t ha⁻¹ in regional potato yield studies (Tamayo-Vera et al., 2025). Random Forest models applied to MODIS NDVI data achieved a correlation of 0.76 in Zambia. Similarly, XGBoost models reduced prediction errors to less than 2 t ha⁻¹ in South Asia (Prodhan et al., 2021) in crops like rice, wheat, and maize. More recently, deep learning techniques such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks have seen increased adoption for crop yield prediction, offering the potential for improved accuracy and performance over traditional models (Kaneko et al., 2019; Kumar et al., 2023; Srivastava et al., 2022). A significant limitation to their widespread application, however, is that they typically require large amounts of data to train effectively, which poses a major technical obstacle in data-scarce regions such as many parts of Africa (Safonova et al., 2023). To mitigate these data limitations, transfer learning has been proposed as a promising solution, enabling models to generalize from smaller training datasets and make robust predictions even under changing climatic conditions (Morales and Villalobos, 2023). Despite this potential, the practical effectiveness of these models—and their transferability across diverse agro-ecological zones and low-data environments—remain critical yet understudied questions (Ma et al., 2023; Stiller et al., 2024).

Addressing the challenge of data scarcity directly, foundation models mark a paradigm shift in deep learning. Introduced in 2021, these models are trained on large and diverse datasets through self-supervised learning and can be adapted to a variety of downstream tasks with minimal additional data (Bommasani et al., 2021). In the tabular domain, TabPFN (Tabular Prior-data Fitted Network) is a recent foundation model capable of few-shot learning for structured data prediction (Hollmann et al., 2025; Hoo et al., 2025). Unlike conventional deep networks that demand extensive training, TabPFN can draw robust inferences from limited samples, offering a transformative solution for yield prediction in data-constrained regions. Nevertheless, its application in agricultural science, particularly for crop yield mapping, remains a critical and largely unexplored research frontier. To the best of our knowledge, this study represents the first empirical application and systematic benchmarking of a tabular foundation model (TabPFN) for crop yield prediction in Africa.

Despite notable progress, a substantial research gap persists at the intersection of high-resolution yield mapping and data-efficient machine learning applications in Africa. Existing datasets —such as

M3Crops, GAEZ, SPAM, and RAY2012—provide valuable continental-scale insights but remain constrained by coarse spatial resolution, discontinuous temporal coverage, and a strong dependence on model-based assumptions (Monfreda et al., 2008; Fischer et al., 2021; Ray et al., 2012; You et al., 2014; Cao et al., 2025). Even more recent hybrid datasets like GDHY, which integrate satellite remote sensing with agricultural census data, remain constrained by their 0.5° spatial resolution (~50 km), limiting their utility for localized agricultural planning (Iizumi and Sakai, 2020). Methodologically, while various machine learning models have been applied, a comprehensive benchmark comparing traditional, deep learning, and emerging techniques in the unique data-scarce context of Africa is still missing. Consequently, there is a pressing need for a scalable framework that achieves policy-relevant spatial resolution by identifying and leveraging the most advanced, data-efficient models available.

To address this gap, this study develops and benchmarks a scalable, data-driven mapping pipeline to support food security in Africa. Our approach is designed to overcome the limitations of existing datasets and to identify the most effective modelling strategies for data-constrained environments.

Accordingly, the main objectives of this study are:

- 1) To develop a high-resolution (~250 m) maize yield prediction framework for Africa by integrating Earth Observation (EO) data (climate and soil variables) with advanced ML models trained on spatially disaggregated synthetic yield data derived from Net Primary Productivity (NPP), to generate high-resolution yield maps.
- 2) To evaluate and compare the performance of multiple ML and deep learning models, including, tree-based ensemble, traditional deep learning in predicting maize yield across African countries, using rigorous time-aware cross-validation techniques, and to assess the effectiveness of tabular foundation models in crop yield prediction.
- 3) To assess the contribution of different environmental and agronomic variables (e.g., climate, vegetation indices, drought indices, soil properties) to maize yield variability based on causal inference techniques.

2. Study area and datasets

2.1. Study area

Africa, the second largest and second-most populous continent in the world, comprises a diverse array of physical, climatic, and ecological zones that make it highly variable in terms of agricultural potential. Maize is one of the most important staple crops across the continent, both in terms of consumption and cultivation. This study focuses on 42 maize-producing countries in Africa, selected based on their contribution to regional maize output, availability of agricultural data, and geographic representation across different agro-ecological zones. The study area extends longitudinally from approximately 17°W to 51°E and latitudinally from 35°N to 35°S, encompassing countries from North Africa (Algeria, Egypt, Libya, Morocco, Sudan, Mauritania), West Africa (Benin, Burkina Faso, Côte d'Ivoire, Gambia, Ghana, Guinea, Mali, Niger, Nigeria, Senegal, Sierra Leone, Togo), Central Africa (Cameroon, Central African Republic, Chad, Congo, Democratic Republic of the Congo) East Africa (Burundi, Eritrea, Ethiopia, Kenya, Rwanda, Somalia, Uganda, United Republic of Tanzania), Southern Africa (Angola, Botswana, Lesotho, Madagascar, Malawi, Mozambique, Namibia, South Africa, Zambia, Zimbabwe). This vast spatial coverage captures the variability in climatic and environmental conditions that influence maize productivity across the continents. The diverse physiography—spanning from low-lying plains like the Congo Basin to high-elevation plateaus and highlands such as the Rift Valley—drives major differences in agricultural patterns. Soil properties are equally varied, with substantial regional differences in texture, pH, organic carbon, and water-holding capacity that directly affect nutrient availability. This

marked climatic and environmental variation is well-represented by the distribution of Köppen climate zones (Fig. 1a).

2.2. Maize yield and crop calendar data

This study utilizes maize yield data from FAOSTAT for the period 2009–2022, aligning with the availability of Net Primary Productivity (NPP) data, which is employed for spatial disaggregation of crop yields (see Section 3.1). To effectively link yield estimates with phenological phases, we incorporate crop calendar information from (Sacks et al., 2010), which provides global gridded data on planting and harvesting periods. This dataset includes key variables such as the start, end, and duration of both planting and harvesting periods. For our analysis, we focused on the primary maize growing season, labelled as the 'major' season in the dataset. The crop calendar raster was clipped and processed to extract country-level planting and harvesting day-of-year (DOY) values across Africa. Given the substantial spatial variability in cropping practices, we aggregated the raster values using the mode to represent the most dominant planting and harvesting dates within each country. This approach ensures consistent and representative

identification of maize growth periods for subsequent analysis with Earth Observation (EO) data.

2.3. Earth observation (EO) datasets

In this study, a range of crop yield determinants was derived from open-source climate, soil, and remote sensing data products (refer to Table 1 for the details).

2.3.1. Climate data

Climate data plays a significant role in crop yield prediction by providing essential information about environmental conditions, including long-term climatic trends and short-term extreme weather events (Chinnasamy and Ashok, 2024). Accordingly, we extracted temperature, precipitation, solar radiation, wind component, and volumetric soil water datasets from the ERA5-Land daily aggregated climate reanalysis dataset, which is hosted on the Google Earth Engine (GEE) platform (Muñoz-Sabater et al., 2021). Reference evapotranspiration (ET_0) also serves as a key indicator of crop water demand and serves as an input to the calculation of drought stress indices, thereby

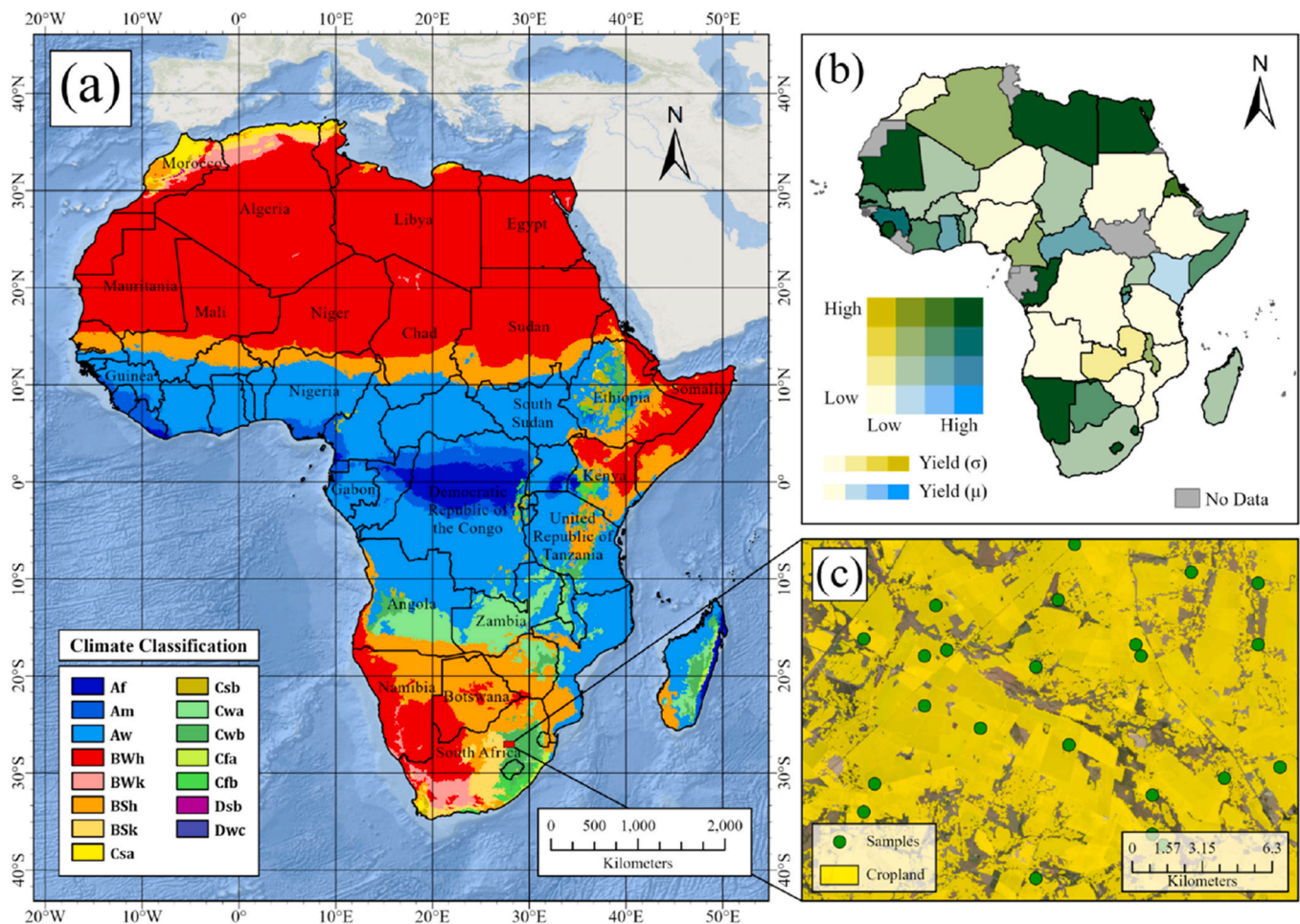


Fig. 1. Overview of the study area and spatial characteristics of historical maize yield across Africa. (a) Köppen–Geiger climate classification map of Africa (Beck et al., 2018) illustrating the continent's wide range of agroecological conditions—from arid desert regions (e.g., BWh in northern Africa) to humid tropical zones (e.g., Af and Aw in central and western Africa). (Climate Classification: Af – Tropical Rainforest Climate, Am – Tropical Monsoon Climate, Aw – Tropical Savanna Climate, BWh – Hot Desert Climate, BWk – Cold Desert Climate, BSh – Hot Semi-Arid Climate, BSk – Cold Semi-Arid Climate, Csa – Hot-Summer Mediterranean Climate, Csb – Warm-Summer Mediterranean Climate, Cwa – Monsoon-Influenced Humid Subtropical Climate, Cwb – Subtropical Highland Climate with Dry Winters, Cfa – Humid Subtropical Climate, Cfb – Oceanic Climate, Dsb – Warm-Summer Continental Climate, and Dwc – Subarctic Climate with Dry Winters), (b) National-level average (μ) and variability (σ) of FAO-reported maize yields (t/ha) for the period 1961–2022. Darker green shades denote higher mean yields, while yellow hues indicate greater interannual variability. Countries shown in grey lacked sufficient data for inclusion. (c) A 10-m spatial resolution cropland extent map from Digital Earth Africa displaying a sample region in southern Africa. Green dots represent training sample points extracted from cropland areas (in yellow), used in constructing the machine learning dataset.

Table 1

Overview of datasets used in the study, including their categories, parameters, units, sources, and spatial resolutions.

Category	Parameter	Unit	Data Source	Spatial Resolution
Climate	Temperature (Tmin, Tmax)	°C	ERA5-Land Daily Aggregated - ECMWF Climate Reanalysis (https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_DAILY_AGGR)	11132 m
	Precipitation (Prcp)	mm day ⁻¹		
	Solar Radiation (Rad)	MJ m ⁻² day ⁻¹		
	Wind Component (Wind)	m s ⁻¹		
	Volumetric Soil Water (VSW)	m ³ m ⁻³		
	Evapotranspiration (ET ₀)	mm day ⁻¹		
Soil	Drought Indicators (SPI-1, SPI-3)	Dimensionless	WAPOR Daily Reference Evapotranspiration 2.0 (https://developers.google.com/earth-engine/datasets/catalog/FAO_WAPOR_2_L1_RET_E) Derived from ERA5-Land Monthly Aggregated	18924 m
	Agro-ecological Zones	Dimensionless	Agro-ecological Zones of Africa (https://doi.org/10.7910/DVN/HJYYTI)	~1 km
	Bulk Density (BDOD)	kg dm ⁻³	Soil Grids 250m v2.0 (https://gee-community-catalog.org/projects/isric/)	250 m
	Cation Exchange Capacity (CEC)	cmol(+)/kg		
	Coarse Fragments (CFVO)	% (volume)		
	Clay Content (Clay)	% (weight)		
	Nitrogen Content (Nitrogen)	g kg ⁻¹		
	Soil pH in Water (pH)	pH units		
	Organic Carbon Density (OCD)	kg dm ⁻³		
	Normalized Difference Vegetation Index (NDVI)	Dimensionless (−1 to 1)	MODIS Terra Daily NDVI (https://developers.google.com/earth-engine/dataset/catalog/MODIS_MOD09GA_006_NDVI)	463 m
Vegetation	Normalized Difference Water Index (NDWI)	Dimensionless (−1 to 1)	MODIS Terra Daily NDWI (https://developers.google.com/earth-engine/dataset/catalog/MODIS_MOD09GA_006_NDVI)	463 m
	Irrigation Data (Irrigation/Irri)	Binary (0 = rainfed, 1 = irrigated)	Landsat-derived Global Rainfed and Irrigated-Cropland Product (LGRIP) (https://gee-community-catalog.org/projects/lgrip30)	30 m
Disaggregation	Cropland	Binary (0 = non-crop, 1 = crop)	Crop planting dates: an analysis of global patterns (Sacks et al., 2010) (https://sage.nelson.wisc.edu/data-and-models/datasets/crop-calendar-datas-et/)	10 m
	Planting and Harvesting dates and time periods	Day of Year (DOY)	Crop planting dates: an analysis of global patterns (Sacks et al., 2010) (https://sage.nelson.wisc.edu/data-and-models/datasets/crop-calendar-datas-et/)	0.5°
	Net Primary Productivity (NPP)	g m ⁻²	WAPOR Dekadal Net Primary Production 2.0 (https://developers.google.com/earth-engine/datasets/catalog/FAO_WAPOR_2_L1_NPP_D)	248 m

contributing to better crop yield estimates (Dong et al., 2024). The ET₀ data utilized in this study were derived from the WAPOR daily reference evapotranspiration dataset, accessible through the GEE platform (Blatchford et al., 2020). The ERA5-Land monthly aggregated dataset was used to derive drought indicators (e.g., SPI-1 and SPI-3).

2.3.2. Vegetation indices

Multi-spectral remote sensing data provides reliable insights into crop biophysical parameters across various growth stages by capturing distinct spectral signatures (Muruganantham et al., 2022). For instance, the Normalized Difference Vegetation Index (NDVI) is widely used to assess crop productivity and identify physiological stress, while the Normalized Difference Water Index (NDWI) provides valuable insights into plant and soil moisture levels. We derived these indices from the MODIS Terra daily NDVI and NDWI datasets to improve the accuracy of maize yield prediction.

2.3.3. Soil and agricultural management data

Soil grids data developed by (Poggio et al., 2021), provided spatial characterization of key soil chemical properties such as sand, silt, clay, pH, nitrogen, and soil organic carbon content etc., to a soil depth of a maximum 2m. These properties are critical for accurate crop yield predictions, as they significantly influence spatial yield variability. Irrigation data helps differentiate between rainfed and irrigated cropland regions, leading to more accurate yield estimates. To include this information in our training dataset, we incorporated the Landsat-derived Global Rainfed and Irrigated-Cropland Product (LGRIP) developed by (P. Teluguntla et al., 2023).

2.3.4. Spatial disaggregation dataset

To spatially limit the sample extraction process from all EO feature layers only to cropland areas, we applied a cropland mask based on the Africa Cropland Extent 2019 dataset from Digital Earth Africa. Lastly,

the disaggregation of crop yield from the country scale to the pixel scale, a unique aspect of this study, is achieved by using Net Primary Productivity (NPP) as a proxy for crop yield at the local scale. NPP quantifies the net primary production of crops, with higher values indicating greater biomass accumulation, which correlates with increased yields and enhances the understanding of agroecosystem dynamics (Zhang et al., 2020). For this purpose, we employed the WAPOR Dekadal Net Primary Production 2.0 dataset, available on the GEE platform, for crop yield disaggregation (FAO, 2018). The crop yield disaggregation methodology is detailed in Section 3, and a comprehensive compilation of all data sources and metadata is provided in Table 1.

3. Materials and methods

In this study, we developed a robust, high-resolution maize yield prediction framework spanning 42 African countries from 2009 to 2022—one of the most spatially and temporally comprehensive efforts to date for the continent. Leveraging 296 features—including dekadal remote sensing and climate time-series data aligned with maize phenology, and high-resolution soil variables—we addressed key data sparsity challenges using a spatial disaggregation method that converts FAO national yield statistics into pixel-level labels via Net Primary Productivity (NPP). While based on prior methods (Iizumi et al., 2013), our approach integrates crop calendars, noise-filtered cropland masks, and quality control thresholds to enhance the reliability and generalizability of disaggregated labels. To ensure temporal rigour, we employed an expanding-window cross-validation strategy and trained multiple machine learning and deep learning models using mean squared error (MSE) loss. Performance was evaluated across both pixel and country scales using multiple metrics. This methodological pipeline enables scalable and interpretable yield prediction, supporting more precise agricultural monitoring across data-constrained regions in Africa. The complete methodological pipeline, including data preprocessing and

model development, is detailed in the following section and illustrated in Fig. 2.

3.1. Spatial disaggregation of FAO yield using NPP-based modelling

The disaggregation process began with spatial sampling across verified cropland areas in Africa. To improve label quality, we applied connected component analysis to the cropland extent mask, removing isolated clusters smaller than 100 pixels that were likely misclassified. This filtering step reduced spatial noise and improved land-use reliability as there was a spatial resolution mismatch between the cropland mask layer (~10m) and the NPP data layer (~250m), which resulted in reduced probability of sampling from land use classes other than croplands. We then calculated cropland extent per country and allocated sample points proportionally, yielding a stratified, spatially balanced training set of ~100,000 points across the continent. To handle data scarcity and ensure a minimum level of statistical robustness, we decided a threshold requiring at least 50 cropland samples per country for inclusion in the model training. Countries with fewer than 50 cropland samples-due to limited agricultural area-were excluded to avoid unreliable model training and statistical artefacts. This threshold represents a critical trade-off between maximizing geographic coverage and ensuring the reliability of the model.

To generate high-resolution training labels, we adapted the yield disaggregation framework of *Iizumi et al. (2013)*. Specifically, we used FAO WaPOR Dekadal Net Primary Production (NPP) version 2.0 as a

proxy to spatially distribute national FAO yield statistics across cropland pixels. NPP was selected as a key metric due to its empirical relationship with net carbon assimilation and the biomass accumulation dynamics fundamental to crop yield formation (*Zhang et al., 2020; Wang et al., 2020*). Unlike spectral indices such as NDVI or LAI, which primarily quantify canopy greenness or leaf area and often suffer from saturation in high-density vegetation or fail to account for respiratory carbon losses, NPP offers a more comprehensive, physiologically meaningful measure of ecosystem productivity. Moreover, the WaPOR NPP product is optimized for agricultural monitoring, providing consistent spatial coverage, quality-controlled cloud-screened composites, and values expressed in physical units (g m^{-2}), which facilitate biologically interpretable allocation of yield. Our approach incorporates additional spatial filtering and temporally aligned crop calendars, ensuring the downscaled yield values capture local phenological and biophysical conditions. Details of the variables used in the spatial disaggregation process are provided in *Table S1.1* and the complete pseudocode is presented in Section S1.2 of the Supplementary Information. The sequential process used to disaggregate national yields into pixel-level estimates is as follows:

- i) First, we obtained the planting and harvest dates, along with the duration of the major maize growth stages, for each country in Africa using the crop calendar dataset developed by (*Sacks et al., 2010*).

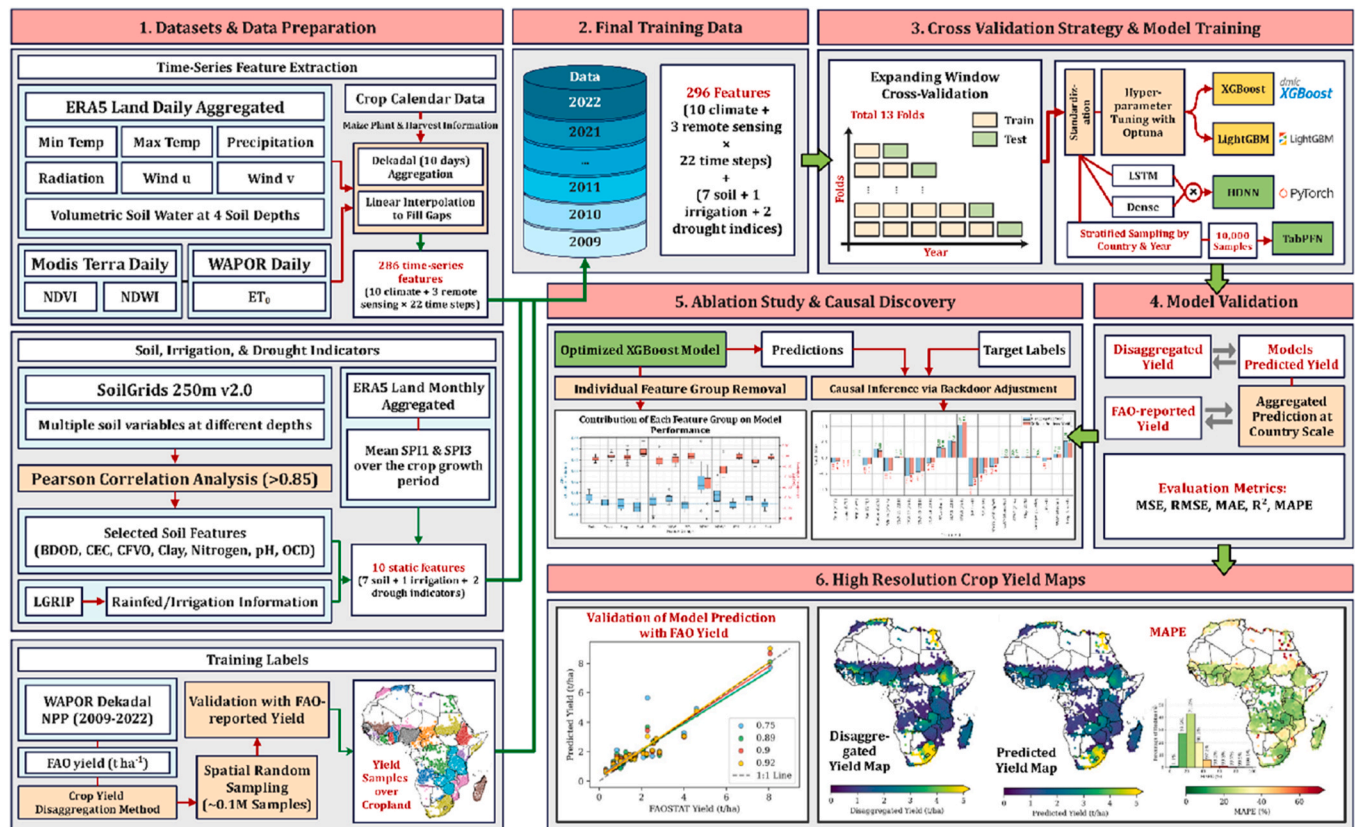


Fig. 2. End-to-end framework for maize yield prediction comprises six rigorously designed components: (1) Multi-source data preprocessing, where climate variables, vegetation indices, irrigation layers, and soil properties are harmonized into phenology-aware time-series and static features; (2) Dataset construction with 296 features- 286 time-series variables aggregated at dekadal resolution and 10 static spatial predictors-aligned to crop growth stages using country-specific planting calendars; (3) Label generation via a novel application of NPP-based disaggregation of FAO national yields to pixel level; (4) Model training using a temporally-expanding cross-validation strategy across four distinct algorithms: tree-based (XGBoost, LightGBM), deep learning (Hybrid Deep Neural Network combining LSTM and dense layers), and TabPFN foundation models; (5) Model evaluation through spatial and country-level metrics (MSE, RMSE, MAE, R^2 , MAPE), ensuring robustness and scalability; and (6) Post-hoc interpretability using both ablation analysis and causal inference (backdoor criterion), culminating in the generation of actionable, high-resolution yield maps. This integrated pipeline bridges explainability, generalization, and fine-scale spatial resolution, enabling practical application in data-scarce agricultural regions.

- ii) Next, a cropland mask was applied to the NPP data product, and dekadal NPP values were aggregated by averaging them temporally over the maize growth period for each country and year. This step produced pixel-level NPP averages while preserving the spatial variation in maize productivity (Fig. 3a):

$$\text{Average NPP}_i = \frac{1}{T} \sum_{t=1}^T \text{NPP}_i(t) \quad (\text{Eq.1})$$

where T represents the total number of dekadal time steps corresponding to the growth periods of a given country, and i represents an individual pixel.

- iii) The country-level NPP average was obtained by spatially averaging the NPP values from the previous step across all cropland pixels within a country for a given year. This average NPP was assumed to correlate with national maize yield as reported in the FAO data (Fig. 3b):

$$\text{Country Average NPP} = \frac{1}{n \times T} \sum_{i=1}^n \sum_{t=1}^T \text{NPP}_i(t) \quad (\text{Eq.2})$$

where n is the total number of cropland pixels in the country.

- iv) The NPP ratio for each pixel was computed by dividing its average NPP by the country's average NPP (Fig. 3c). This ratio represents whether a pixel exhibits higher or lower productivity compared to the national average.

$$\text{NPP Ratio}_i = \frac{\text{Average NPP}_i}{\text{Country Average NPP}} \quad (\text{Eq.3})$$

- v) The FAO-reported national yield was used as a uniform value across all cropland pixels within a country for a given year. The spatial distribution of yields was derived by multiplying the FAO yield by the corresponding NPP ratio (Fig. 3d–f):

$$\text{Disaggregated Yield}_i = \text{FAO Uniform Yield per Pixel} \times \text{NPP Ratio}_i \quad (\text{Eq.4})$$

This methodology enables the disaggregation of country-level yield data to a finer spatial resolution to capture spatial heterogeneity in crop yield predictions—an aspect often lacking in previous studies. However, to ensure the reliability of the disaggregated yield estimates before their use in subsequent modelling steps, we re-aggregated the data to the national scale and compared it with FAO yields. To maintain data quality, countries with a mean absolute percentage error (MAPE) exceeding 5 % between the re-aggregated and FAO yields were excluded. After applying these post-processing steps, yield data for approximately 42 countries were retained for further model development. The entire yield disaggregation process is visually represented in Fig. 3.

3.2. Data preparation for machine learning model training

Data preparation is a significant and sensitive step in our modelling framework, as the length of the growth period varies across countries. To account for these variations, we processed the data iteratively for each country, adjusting the planting and harvest dates accordingly (refer to section 2.2 for details). Time-series variables (e.g., climate and remote sensing data) were extracted at a decadal interval from 2009 to 2022,

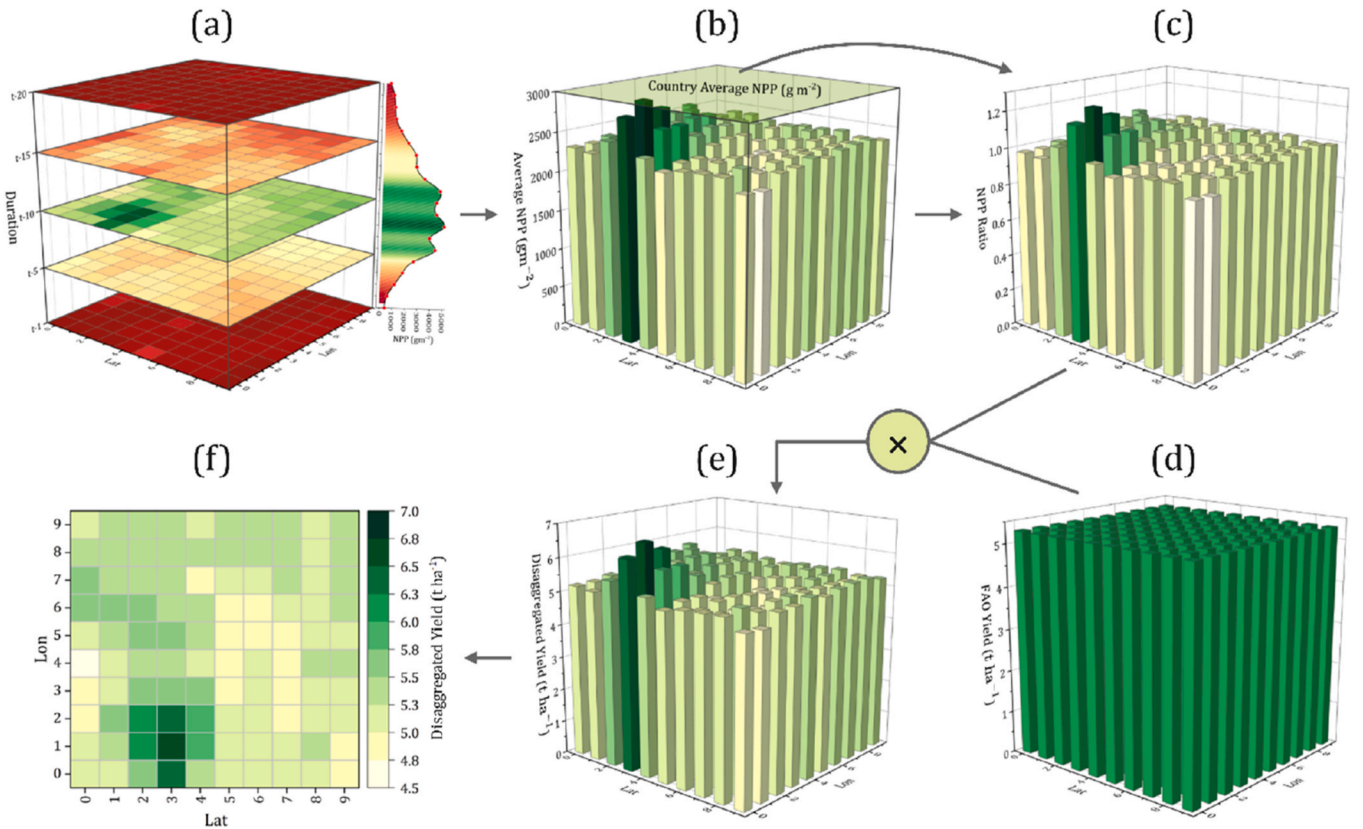


Fig. 3. Schematic representation of the crop yield disaggregation method adopted from (Iizumi et al., 2013). (a) Temporally aggregated NPP data over the maize growing season using country-specific planting and harvest dates from the crop calendar dataset by Sacks et al. (2010); (b) country-level average NPP calculated from cropland pixels; (c) NPP ratio computed as the ratio of each pixel's average NPP to the country average; (d) uniform FAO national yield applied to all cropland pixel; (e) pixel-level disaggregated yield estimated by multiplying the FAO yield with the NPP ratio; (f) resulting spatial distribution of maize yield estimates.

aligning with the availability of NPP data. The spatially disaggregated yield estimates, derived from the NPP data, served as training labels for our model.

To incorporate edaphic factors, we extracted high-resolution soil attributes from the SoilGrids 250m v2.0 database at all sampled locations. Given the multivariate nature of the dataset across various depths, we conducted a Pearson correlation analysis to eliminate multicollinearity and improve model efficiency. We removed 52 highly correlated variables ($r > 0.85$) and retained seven non-redundant soil predictors with agronomic relevance, including bulk density, cation exchange capacity, clay content, and nitrogen concentration (refer to Table 1). This feature selection was essential for ensuring both computational tractability and interpretability of the model across heterogeneous soil environments.

Time-series features—including NDVI, NDWI, and reference evapotranspiration (ET_0)—were compiled at dekadal resolution, covering up to 22 intervals based on the longest observed maize growth cycle across Africa. Missing values, primarily due to remote sensing artefacts, were imputed using linear interpolation to maintain temporal continuity. To enhance the representation of climate extremes, we computed Standardized Precipitation Indices (SPI-1 and SPI-3) over each country's crop growth period, capturing short- and medium-term drought dynamics. We further incorporated irrigation extent from the LGRIP global database, allowing the models to distinguish between rainfed and irrigated conditions—an often overlooked yet critical factor in large-scale yield prediction (refer to Table 1).

3.3. Cross-validation and model optimization

A robust and generalizable yield prediction framework was developed by evaluating four complementary modelling approaches, encompassing tree-based ensemble methods, deep neural networks, and transformer-based probabilistic models. This diverse set of models (detailed in Section 3.4) captures both temporal dynamics and spatial variability, with particular emphasis on the novel inclusion of TabPFN—a pretrained foundation model offering efficient probabilistic inference for tabular data. Hyperparameter tuning for the ensemble models was conducted using the Optuna framework with Bayesian optimization (Akiba et al., 2019).

To rigorously optimize model performance and prevent overfitting,

we employed two distinct yet complementary cross-validation approaches. The first evaluates the model's ability to predict into the future (temporal generalization), while the second tests its ability to predict for entirely unseen geographic regions (spatial generalization).

We first implemented an expanding window temporal cross-validation strategy to simulate a realistic prediction scenario. This method fully respects the chronological order of the data. The evaluation begins with a model trained on the earliest year of data (2009) and tested on the subsequent year (2010). In each successive fold, the training window expands to include one more year of historical data, while the test set always consists of the single, immediately following year. For example, Fold 1 trains in 2009 and tests in 2010; Fold 2 includes 2009–2010 for training and 2011 for testing, continuing until the final fold evaluates 2022. This setup allows us to rigorously assess model performance under realistic, year-ahead forecasting conditions and inter-temporal yield variability. The full temporal cross-validation strategy is depicted in Fig. 4.

We further extended the expanding-window temporal cross-validation by introducing a spatial hold-out component to assess the framework's geographic generalization capability. In this setup, the model was tested on entirely unseen regions to simulate realistic out-of-domain prediction scenarios. Data were grouped by agroecological zones (AEZs), from which approximately 40 % of the zones were randomly selected and held out as the test set in each fold, ensuring that the test samples included at least 30,000 data points. The model was trained on the remaining AEZs across all available years, and the process was repeated iteratively for each temporal split as the training window expanded over time. This spatio-temporal cross-validation design provided a rigorous assessment of the model's robustness in predicting crop yields across both time and space, particularly in regions characterized by distinct climatic and edaphic conditions.

Machine learning and deep learning models often require input variables to be within the same range, necessitating additional feature scaling steps during cross-validation process. For each fold, the mean and standard deviation of the training data were calculated, excluding no data values associated with extra timesteps. These computed parameters were then applied consistently to both the training and test datasets to scale the input features. Furthermore, while XGBoost, LightGBM, and TabPFN automatically handle no-data values, the HDNN model does not. To address this inconsistency, missing data were

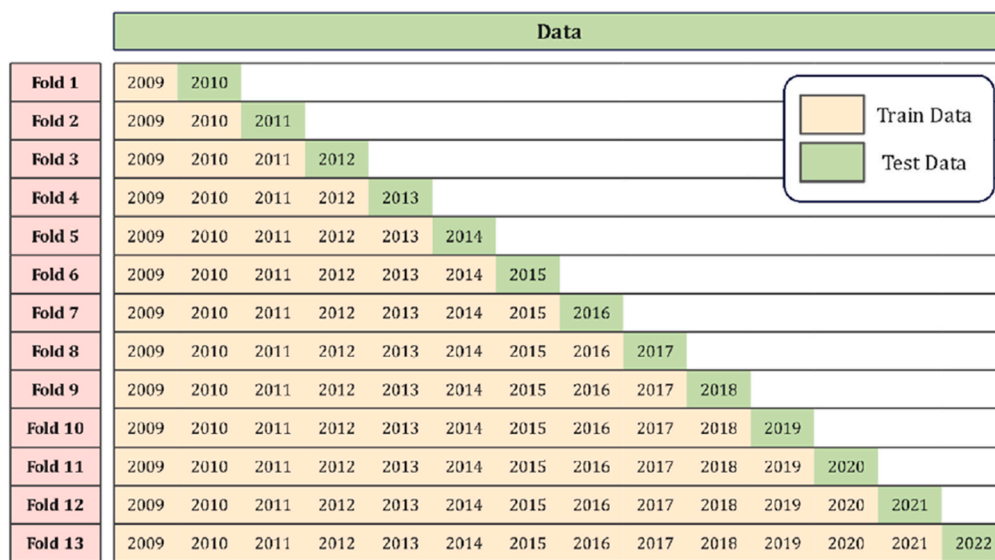


Fig. 4. Schematic representation of the expanding window cross-validation strategy used to evaluate model robustness and generalizability. Each fold incrementally increases the size of the training window by one year and validates performance on the subsequent year, starting from Fold 1 (training in 2009, testing on 2010) and progressing until Fold 13 (training in 2009–2021, testing on 2022). The beige cells denote the training years, while the green cells indicate the corresponding test year for each fold.

imputed with a neutral value of zero before training the HDNN model.

The Optuna framework requires an objective value to be minimized during the training process. We used the mean MSE across all folds as the final loss criterion to optimize machine learning model predictions. This process trains a new model for each fold and validates predictions based on yield values from the subsequent year. We conducted 30 optimization trials, with each trial involving the training of 13 models to determine the optimal hyperparameters for both machine learning models. For deep learning models, HDNN and TabPFN, we adopted the same data-splitting strategy, training a new model from scratch at each fold. Additionally, for the HDNN model, the epoch limit was set to 50 and early stopping was implemented to stop training if the test loss did not decrease for five consecutive epochs. Since the current version of TabPFN supports a maximum of 10,000 data points and 500 features, we applied stratified sampling based on country and year to ensure a representative subset, as the number of data points increased with each fold in the actual dataset. A particularly noteworthy aspect, discussed in the following section, is the performance of TabPFN. Despite being trained on a significantly smaller dataset ($\leq 10,000$ data points), it achieves accuracy comparable to XGBoost, LightGBM, and HDNN, which require substantially larger training datasets. A detailed description of the models is provided in the following section.

3.4. Overview of TabPFN and benchmark models

To benchmark model performance, we employed three well-established machine learning algorithms—XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017), and a Hybrid Deep Neural Network (HDNN)—alongside the recently developed Tabular Prior-data Fitted Network (TabPFN) (Hollmann et al., 2025). TabPFN represents a fundamental shift in the paradigm of tabular data modelling. To elucidate the practical novelty of TabPFN, it is instructive to contrast its methodological framework with that of conventional machine learning models such as XGBoost and LightGBM. The key distinction lies in both the training paradigm and the inference process. Unlike conventional models that start as a "blank slate" and must be trained extensively on a specific, real-world dataset, TabPFN is a pre-trained "generalist." It has already been exposed to a vast array of synthetic datasets generated through structural causal models (SCMs). This pre-training enables it to internalize broad principles of pattern recognition within tabular data.

This foundation gives rise to a fundamentally different approach to inference. Whereas conventional models follow a distinct two-step "train-then-predict" workflow, TabPFN effectively collapses these two stages into a single forward pass using a technique called in-context learning (ICL). During inference, TabPFN is provided with the entire dataset—both training and test samples—simultaneously. It uses the training data as "in-context examples" to instantly recognize the pattern for the specific task and applies it to the test data within the same operation. TabPFN mimics Bayesian inference and can do both training and prediction in just one forward pass through a transformer-based neural network. Specifically, given a training set $D_{\text{train}} = \{(x_i^{\text{train}}, y_i^{\text{train}})\}_{i=1}^n$ and a test set $D_{\text{test}} = \{(x_j^{\text{test}})\}_{j=1}^m$, the model approximates the posterior predictive distribution:

$$p(\widehat{y}^{\text{test}} | X^{\text{test}}, X^{\text{train}}, Y^{\text{train}}). \quad (\text{Eq.11})$$

This distribution is directly produced by a neural network:

$$\widehat{y}^{\text{test}} = f_{\theta}(X^{\text{train}}, Y^{\text{train}}, X^{\text{test}}) \quad (\text{Eq.12})$$

where f_{θ} represents the TabPFN transformer, which has been trained on millions of synthetically generated datasets.

TabPFN adapts the traditional transformer encoder architecture to better suit the characteristics of tabular data. In this design, each cell in the table is treated as a separate token, and the model employs a two-dimensional attention mechanism—one along the feature dimension

(across columns) and the other along the sample dimension (across rows). This structure ensures that the model remains invariant to the ordering of both features and samples, which is crucial for effectively handling tabular data. After each attention block, a feed-forward multilayer perceptron (MLP) is applied, along with standard residual connections and layer normalization.

From a regression perspective, instead of directly predicting point estimates for target values, TabPFN outputs a full predictive distribution over possible target values for each test sample. This probabilistic approach allows TabPFN to naturally express uncertainty and support multimodal regression tasks. The model represents the target variable as a piecewise constant probability distribution over predefined interval $\{I_k\}_{k=1}^K$. For each test sample, it predicts a probability vector $\pi = (\pi_1, \dots, \pi_K)$, representing the likelihood of the target falling within each interval. The resulting predictive density is given by

$$\widehat{p}(y) = \sum_{k=1}^K \pi_k \cdot I[y \in I_k] \quad (\text{Eq.13})$$

The loss function is the negative log-likelihood of the true target value under this distribution. If the true value y lies in interval I_k , the loss becomes

$$L_{\text{regression}} = -\log \pi_k \quad (\text{Eq.14})$$

This formulation enables TabPFN to learn not only point predictions but also the full distributional structure, capturing uncertainty and multimodality in target values.

Detailed mathematical formulations and training configurations of the benchmark models (XGBoost, LightGBM, and HDNN) are provided in Supplementary Section S2.

3.5. Ablation study and causal discovery

To rigorously assess the influence of different environmental and management variables on yield prediction accuracy, we conducted a structured ablation study on the XGBoost model. Unlike generic feature importance methods, our approach uniquely groups input variables into interpretable thematic categories—soil properties, minimum and maximum temperatures, precipitation, solar radiation, volumetric soil water content, drought indices, reference evapotranspiration (ET_0), NDVI, NDWI, and irrigation—enabling a domain-informed evaluation of their collective contribution. Each group was systematically removed in isolation to quantify its impact on predictive performance, providing actionable insights into which biophysical signals are most critical for accurate yield estimation.

To ensure methodological rigour, all ablation experiments were conducted using an identical set of hyperparameters optimized via the Optuna framework and evaluated under the same expanding window cross-validation protocol as the baseline model. This setup yielded 12 controlled training scenarios, each representing the absence of one feature group. Resulting outputs were stored and benchmarked against the full model to determine the marginal loss in accuracy attributable to each exclusion. This systematic evaluation enhances the interpretability and trustworthiness of the model, offering valuable guidance for both scientific understanding and feature selection in operational prediction systems.

To go beyond mere correlation and assess whether the machine learning model captured true causal signals in the data, we implemented a robust causal discovery framework grounded in structural causal theory. Specifically, we applied a backdoor-adjustment-based inference approach to uncover the underlying cause-and-effect relationships between key environmental drivers—such as climate variables, soil properties, and vegetation indices—and both the observed (disaggregated) and predicted crop yields. This dual-path analysis allowed us to not only estimate real-world causal effects but also to test the fidelity of the XGBoost model in preserving these causal structures, a novel and critical

step in validating model interpretability and trustworthiness.

Guided by domain expertise and exploratory analysis, we identified appropriate confounding variable sets for each treatment variable, then constructed two parallel causal graphs using the DoWhy library (<https://www.pywhy.org/dowhy/v0.12/>): one targeting the actual yield and the other using the XGBoost-predicted yield as the outcome. By adjusting for confounders satisfying the backdoor criterion, we computed the Average Treatment Effect (ATE) to isolate the marginal causal influence of each treatment. This rigorous approach enables a more principled evaluation of model alignment with real-world agromonic processes, enhancing both the scientific robustness and the practical utility of data-driven decision tools in agriculture.

For a given treatment T , we adjusted for a set of confounders Z that satisfy the backdoor criterion—i.e., they block all non-causal paths from T to crop yield Y in the causal graph. The causal effect of interest is quantified as the average treatment effect (ATE):

$$\text{ATE} = E[Y|do(T=1)] - E[Y|do(T=0)] \quad (\text{Eq.15})$$

The ATE quantifies the expected change in the outcome (e.g., yield) if we change the treatment from 0 to 1 for every data point in the population. The interventional quantity can be estimated from observational data by applying the backdoor adjustment formula:

$$E[Y|do(T=t)] = \sum_z E[Y|T=t, Z=z] \cdot P(Z=z) \quad (\text{Eq.16})$$

To estimate the causal effect, we fit a linear regression model of the form $Y = \beta_0 + \beta_1 T + \beta_2^T Z + \epsilon$, where the coefficient β_1 represents the estimated causal impact of T on Y . This analysis is conducted for both the observed yields and the XGBoost-predicted yields to assess whether the predictive model preserves the underlying causal relationships.

3.6. Quantitative assessment of regression models

Commonly used validation metrics for regression tasks—particularly in crop yield prediction—include mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), coefficient of determination (R^2), and mean absolute percentage error (MAPE) (Deshmukh et al., 2022). These metrics are essential for evaluating both the performance and accuracy of the predictive models developed in this study. While conceptually related, each metric provides unique insights into different aspects of model behavior and error characteristics.

MSE calculates the average squared difference between predicted and actual yield values, placing greater importance on larger errors, however, it is less interpretable due to its squared units. RMSE, the square root of MSE, offers a more interpretable metric by retaining the same units as the original data, but it remains sensitive to outliers. MAE computes the average of absolute differences between predicted and actual values, providing a more intuitive measure and being robust to outliers, though it does not penalize large errors as severely.

R^2 represents the proportion of variance in the actual data that is explained by the model. While it indicates the model's ability to fit the data, it does not always reflect prediction accuracy and can be misleading, particularly in cases of overfitting. MAPE expresses errors as percentages, making it useful for comparing models across different spatial and temporal contexts. However, it requires caution, as small actual values (approaching zero) can result in inflated error rates.

The mathematical interpretation of each of these metrics is as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (\text{Eq. 17})$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{Eq. 18})$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (\text{Eq. 19})$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (\text{Eq. 20})$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (\text{Eq. 21})$$

In the context of our study, y_i represents the disaggregated yield, $\hat{y}_i$ denotes the predicted yield, n is the total number of observations, and $\bar{y}$ is the mean of the disaggregated yield values.

All models were implemented in a Python v3.12 environment using XGBoost v3, LightGBM v4, and PyTorch v2.2.0. TabPFN was applied via its official GitHub implementation. Hyperparameter tuning for XGBoost and LightGBM was performed using Optuna v4, with 30 independent trials and Tree-structured Parzen Estimator (TPE) samplers optimizing parameters such as learning rate, tree depth, and regularization. Experiments were run on a high-performance computing cluster equipped with NVIDIA Tesla V100 GPUs, while TabPFN utilized NVIDIA H100 GPUs. The full codebase is publicly available at the GitHub repository.

4. Results

4.1. Comparative performance of models across cross-validation folds

Fig. 5 presents a year-by-year evaluation of model performance using disaggregated yield data as the target variable. Metrics were calculated across all sample points for each test year, providing temporal insights into the models' capacity to capture spatial yield variability. The top panel illustrates the coefficient of determination (R^2) from 2010 to 2022. XGBoost consistently achieved high R^2 scores, peaking at 0.87 in 2012 and 0.84 in 2013, underscoring its robustness across diverse conditions. LightGBM and HDNN performed comparably, often closely trailing XGBoost. TabPFN, while operating on significantly smaller datasets, exhibited competitive R^2 in some years, though it underperformed between 2019 and 2021—possibly reflecting challenges in modelling recent variability. Average R^2 values across the full period reinforce these trends, positioning XGBoost (0.78) slightly ahead of LightGBM (0.77), HDNN (0.76), and TabPFN (0.75), thereby highlighting its strong generalization ability.

The bottom panel of Fig. 5 presents the Mean Absolute Percentage Error (MAPE) by year, providing a complementary view of relative prediction accuracy. TabPFN records the lowest average MAPE (25.07 %), followed closely by LightGBM (25.44 %) and XGBoost (25.53 %). Interestingly, although XGBoost leads in R^2 , its slightly higher MAPE suggests greater vulnerability to relative errors in certain instances. This divergence underscores the importance of evaluating both absolute and relative performance. The pronounced variability in MAPE values during 2019 and 2021 hints at increased noise or instability in the data, potentially driven by climate anomalies or inconsistencies in remote sensing observations during those years.

To test spatial generalizability of the modelling framework, a spatial hold-out cross-validation was conducted on XGBoost and TabPFN, revealing distinct geographic patterns in predictive performance across Africa's agro-ecological zones (AEZs) (Fig. 6). Overall, XGBoost demonstrated greater robustness in arid and semiarid environments—such as the Sahelian belt (Mali, Niger, Chad, and Sudan) and the southern African drylands (Namibia, Botswana, and Zimbabwe)—achieving MAPE values around 42–43 %, compared to higher errors for TabPFN (up to 47 %) in these regions. However, these zones are of limited agricultural importance due to their predominantly desert

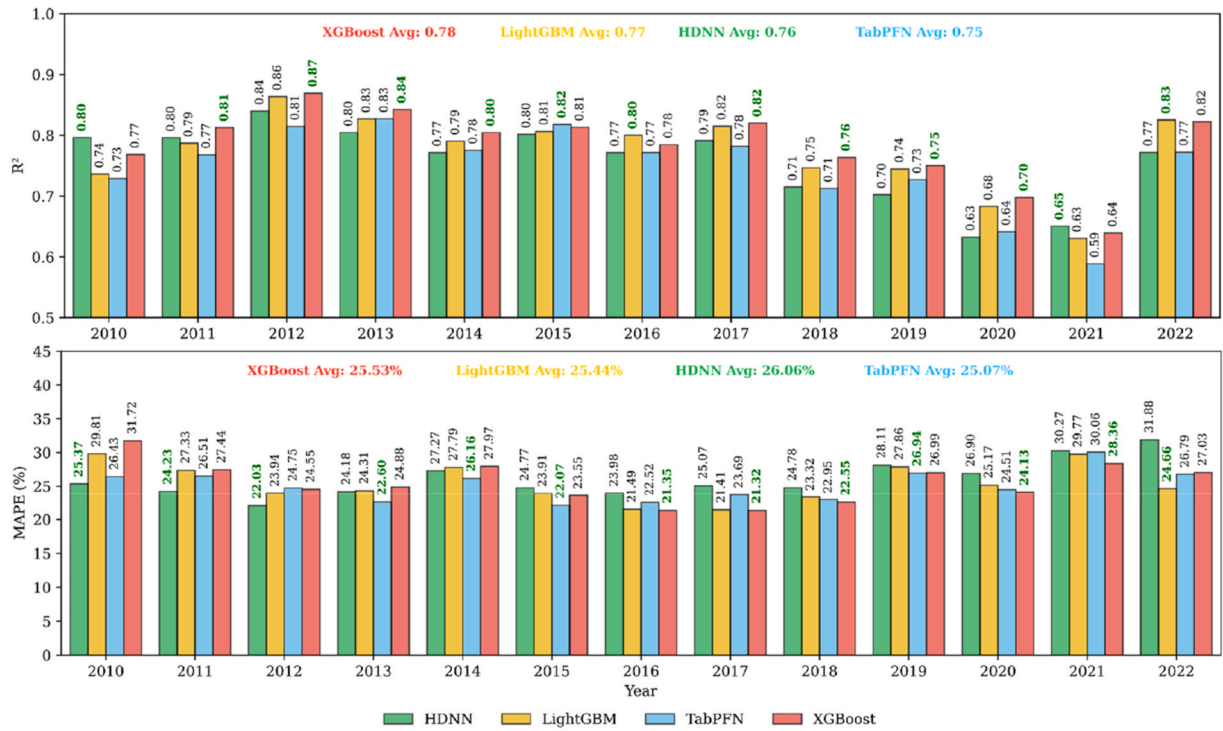


Fig. 5. Year-wise comparison of four models (e.g., HDNN, LightGBM, TabPFN, and XGBoost; see section 3.4 for model descriptions) performances for maize yield prediction using disaggregated yield data as training targets. The top panel shows model explanatory power (R^2), while the bottom panel reflects relative prediction errors (MAPE) highlighting interannual variations in model performance and the ability of each model to capture spatial yield patterns across time.

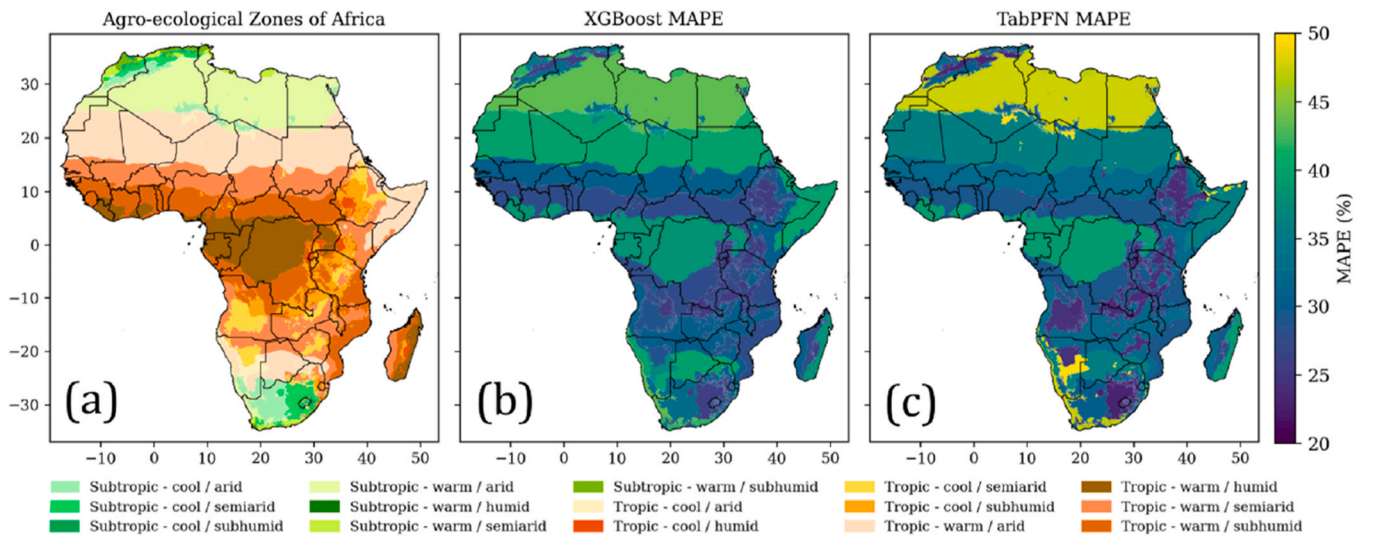


Fig. 6. Spatial hold-out cross-validation results showing the Mean Absolute Percentage Error (MAPE) of XGBoost and TabPFN models across Africa's agro-ecological zones (AEZs).

landscapes and minimal cropland presence. In contrast, TabPFN achieved lower errors in cooler and subhumid regions, including the Ethiopian and Kenyan highlands, the Rwandan plateau, and South Africa's Highveld, with MAPE values ranging between 23 and 30 %. This indicates its superior performance in agriculturally active areas characterized by complex biophysical interactions and stable vegetation patterns. Both models showed comparable accuracy in humid regions such as coastal West Africa (Ghana, Côte d'Ivoire, and Nigeria) and eastern Mozambique, where MAPE values generally remained below 30 %. Notably, TabPFN underperformed in the tropic cool/arid zone (e.g., the dry Ethiopian highlands), suggesting limited spatial generalization

in data-scarce or topographically heterogeneous environments. Furthermore, the observed spatial error patterns may also reflect differences in the density of sample points and the proportion of cropland across regions, with lower errors typically occurring in areas where cropland coverage and data availability were higher.

Table 2 provides a more comprehensive and generalized evaluation of model performance by aggregating key metrics—MSE, RMSE, MAE, R^2 , and MAPE—across all years. It offers a holistic view by combining results from individual folds, allowing for a clearer comparison of each model's overall accuracy and consistency throughout the prediction period. Among the models, there is a trend that XGBoost is more

Table 2

Summary of model performance aggregated over all cross-validation folds, evaluated using multiple error metrics.

Model	MSE (t ha^{-1}) ²	RMSE (t ha^{-1})	MAE (t ha^{-1})	R ²	MAPE (%)
XGBoost	0.57 (± 0.33)	0.73 (± 0.20)	0.47 (± 0.09)	0.78 (± 0.06)	0.26 (± 0.03)
LightGBM	0.59 (± 0.34)	0.75 (± 0.20)	0.49 (± 0.10)	0.77 (± 0.06)	0.25 (± 0.03)
HDNN	0.64 (± 0.35)	0.77 (± 0.20)	0.50 (± 0.11)	0.76 (± 0.06)	0.26 (± 0.03)
TabPFN	0.66 (± 0.38)	0.79 (± 0.21)	0.49 (± 0.10)	0.75 (± 0.07)	0.25 (± 0.02)

accurate with respect to absolute error-based metrics, having the lowest average MSE (0.57), RMSE (0.73), and MAE (0.47). While TabPFN records the lowest average MAPE (25.07 %), its comparatively lower R² (0.75) and higher MSE (0.66) and RMSE (0.79) suggest limitations in its generalizability across varying temporal splits, especially when compared to tree-based models such as XGBoost and LightGBM.

Beyond average performance, the standard deviation ($\pm$) associated with each metric offers valuable insights into the stability and robustness of the models under an expanding window evaluation framework. A lower standard deviation implies more consistent performance across different temporal splits. In this regard, XGBoost demonstrates strong reliability, with relatively low variability in RMSE (± 0.20), MAE (± 0.09), and R² (± 0.06). Although TabPFN shows a slightly lower standard deviation in MAPE (± 0.02), it exhibits the highest variability in R² (± 0.07) and MSE (± 0.38), indicating less consistent behavior over time, which is a potential limitation in operational prediction. HDNN and LightGBM also show moderately higher variances in MSE and MAE compared to XGBoost, which may reflect their sensitivity to temporal fluctuations or varying data distributions.

Overall, when considering both average performance and standard deviation, XGBoost emerges as the most balanced and reliable model, offering a favourable trade-off between accuracy and consistency. LightGBM follows closely behind, while HDNN and TabPFN -despite competitive performance in specific metrics -may benefit from further tuning to enhance robustness under temporally dynamic conditions.

4.2. Comparative performance of models across spatial variations

Fig. 7 provides a spatial performance evaluation of the four predictive models across Africa, using disaggregated yield data as the benchmark. For each model, the actual yields (Fig. 7a, d, 7g, 7j) are compared with corresponding predictions (Fig. 7b, e, 7h, 7k), while MAPE heatmaps reveal spatial patterns of predictive error. To facilitate meaningful regional comparison, particularly in densely sampled zones-spatial aggregation was implemented via hex binning, with MAPE computed for each hexagon using all test-year samples within its boundary. A distinct spatial gradient in model performance is evident. Regions with denser training data coverage-such as Southern Africa (e.g., South Africa, Zambia), East Africa (e.g., Kenya, Ethiopia), and the maize belt in West Africa (e.g., Nigeria, Ghana)-consistently show lower MAPE values, indicating stronger spatial generalization. These areas likely benefit from richer training data and more predictable agro-climatic conditions. In contrast, higher MAPE values are concentrated in data-scarce or environmentally unstable regions, including the Sahel, the Horn of Africa (e.g., Somalia), and parts of Central Africa, where lower sampling density and climatic variability reduce predictive accuracy. Notably, while XGBoost achieves the highest average R² overall, its spatial error distribution is less uniform. The MAPE map (Fig. 7c) shows substantial prediction errors in Eastern Africa, semi-arid zones of the Sahel, and countries such as Zimbabwe and Mozambique. LightGBM and HDNN follow similar spatial trends, though LightGBM marginally outperforms in parts of East Africa. In contrast, HDNN reveals pronounced under-performance across broad regions, including the Horn, Sahel, and parts of Central and Western Africa-highlighting the model's reduced ability to generalize across diverse climatic and agronomic environments. These findings underscore the importance of spatially explicit evaluation and suggest that model robustness is intricately linked to training

data density and regional environmental heterogeneity.

4.3. Performance evaluation against FAO yield statistics

To evaluate the real-world validity of our modelling framework, we compared predicted yields against official national statistics from FAOSTAT. While prior sections assessed accuracy using synthetically disaggregated training labels (i.e., disaggregated yield data from NPP), this comparison anchors model performance in empirical ground truth, offering an external benchmark for assessing operational readiness. We focused on two representative models: XGBoost, due to its consistently strong cross-validation results, and TabPFN, which demonstrated superior spatial generalizability despite being trained on significantly fewer data points.

Fig. 8a and b visualize the relationship between model predictions and FAOSTAT-reported yields at the annual national level. Predictions were aggregated to the country's scale and evaluated year by year. Both models exhibit strong alignment with reported values: XGBoost achieves an R² of 0.79 and a MAPE of 31.8 %, while TabPFN records a slightly lower R² (0.78) but notably better relative accuracy (MAPE = 28.7 %). Results are visualized using hexagonal binning to reduce overplotting and capture local error variability. While XGBoost explains more variance overall, TabPFN's lower relative error indicates higher stability in proportionate yield estimation, which is particularly valuable in policy and decision-support contexts.

Fig. 8c and d provide year-wise performance dynamics, revealing insights into temporal consistency. XGBoost consistently delivers R² > 0.80, peaking at 0.87 (2013), 0.86 (2018), and 0.90 (2022). Notably, TabPFN- despite its smaller training set-either matches or outperforms XGBoost from 2011 to 2016, demonstrating the scalability of foundation models in low-data regimes. A performance dip observed from 2019 to 2021 in both models may reflect real-world yield volatility, anomalous climate shocks, or inconsistencies in FAO reporting. Nevertheless, TabPFN maintains lower or comparable MAPE across most years, and even exceeds XGBoost by 5–10 % between 2014 and 2016 in relative error terms.

Overall, these findings validate the transferability of sub-national disaggregated training labels to national-scale predictions, reinforcing the framework's robustness and applicability. The combination of high explanatory power and relative error consistency highlights the novel contribution of our dual-model approach, offering both accuracy and reliability in predicting maize yields across diverse agroecological zones and periods.

4.4. Spatial distribution of prediction error at the national scale

To evaluate the spatial variability of model performance at the national scale, we calculated the MAPE between the country-level aggregates of predicted yield and compared them with the corresponding FAO yield data. Fig. 9 illustrates the spatial distribution of MAPE values for the XGBoost (Fig. 9a) and TabPFN (Fig. 9b) models, averaged across the test years from 2010 to 2022. Overall, both models achieved low to moderate MAPE across most African countries, although country-specific differences in prediction accuracy were evident.

The XGBoost model demonstrated generally strong performance across the continent, with around 80 % of countries recording MAPE values below 40 %. However, the accuracy of predictions varied

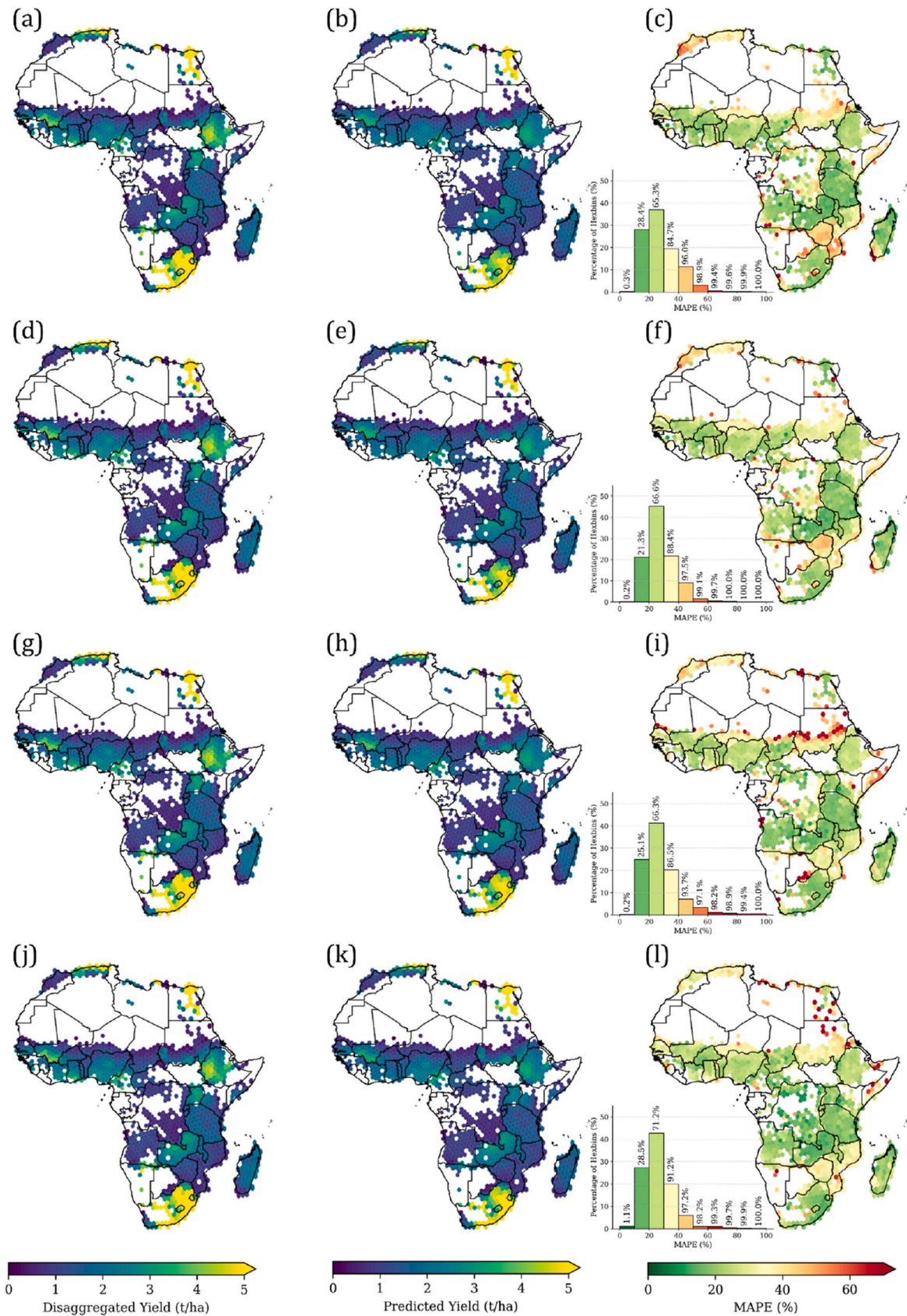


Fig. 7. Spatial evaluation of four predictive models across Africa using disaggregated maize yield data as a reference. Panels (a, d, g, j) display the same disaggregated yield for easy visual comparison, while panels (b, e, h, k) present the predictions from XGBoost, LightGBM, HDNN, and TabPFN. The corresponding MAPE maps (c, f, i, l) show spatial variations in prediction accuracy. Hex binning is applied for spatial aggregation, with MAPE values calculated per hexbin to facilitate meaningful regional comparisons. Histograms in each MAPE panel illustrate the distribution of hexagonal bins across specific MAPE ranges (e.g., 0–10 %, 10–20 %, etc.), with the cumulative percentage of hexagons annotated on each bar.

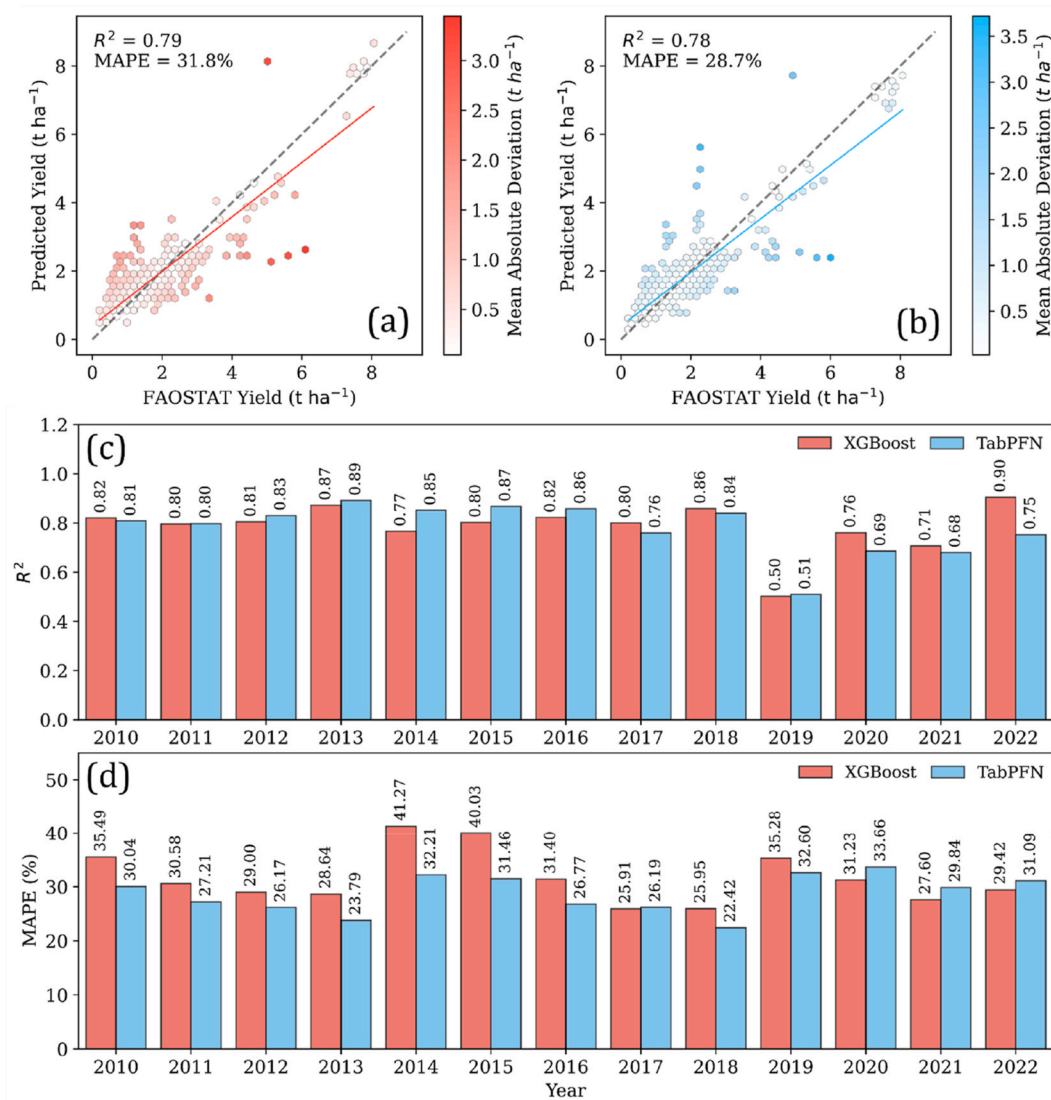


Fig. 8. Comparison of model-predicted (trained following the cross-validation strategy as shown in Fig. 4) aggregated yields with FAO-reported yields at the national scale for maize across Africa. (a, b) Hexagonal bin scatterplots showing the relationship between predicted and FAO yields for XGBoost and TabPFN, respectively. Color gradients represent the mean absolute deviation within each hexbin. (c) Year-wise R^2 values from 2010 to 2022, illustrating the models' ability to explain interannual yield variability. (d) Year-wise MAPE values, indicating the relative prediction errors over cross-validation folds. While XGBoost consistently achieves high R^2 scores, TabPFN demonstrates lower relative errors and competitive performance, particularly in earlier years. Both models exhibit a decline in performance between 2019 and 2021, possibly due to increased yield variability, climate anomalies, or discrepancies in the FAO-reported data.

considerably by region. In North Africa, countries such as Morocco, Algeria, and Niger, as well as Sudan in the Sahel, showed moderate to high error levels ranging from 30 % to 60 %, reflecting a moderate alignment between model predictions and official FAO yield data due to the region's diverse agro-climatic conditions. In West Africa, the model performed particularly well. Key maize-producing countries like Nigeria and Ghana, alongside Mali, Burkina Faso, Guinea, Mauritania, and Senegal, achieved low MAPE values between 10 % and 30 %, indicating high prediction accuracy. An exception in this region was Sierra Leone, where MAPE exceeded 50 %. Model performance in East Africa was more variable. Countries such as Kenya, Uganda, Tanzania, and Malawi recorded low errors (10–20 %), while Ethiopia exhibited slightly higher MAPE values (20–30 %). Somalia was a notable outlier, where the model failed to capture yield patterns, resulting in extremely high error rates. Southern Africa displayed the greatest spatial variability. Countries including Zimbabwe, Botswana, Lesotho, and Eswatini experienced relatively poor model performance, with high MAPE values indicating substantial prediction challenges. In contrast, South Africa-the largest

maize producer in the region, along with Zambia and Madagascar, recorded MAPE values below 20 %, highlighting the model's strong predictive ability in those regions.

The TabPFN model produced a broadly similar spatial pattern to XGBoost, but with 73.2 % of countries achieving MAPE values below 40 %. Notably, TabPFN outperformed XGBoost in several countries where the latter showed higher errors, including the Democratic Republic of Congo, Zimbabwe, Botswana, Lesotho, and Somalia, indicating its strength in capturing yield variability in more challenging or data-sparse regions. However, the model exhibited slightly higher errors in parts of North and Northwest Africa, particularly in countries such as Niger, Morocco, Mauritania, Mali, and Libya. In East Africa, TabPFN demonstrated a marked improvement in prediction accuracy, reducing MAPE in Kenya by approximately 10 %, underscoring its potential for enhanced performance in certain regional contexts.

While both XGBoost and TabPFN demonstrated strong predictive performance at the national scale across Africa, XGBoost exhibited a slight advantage, with an additional 6 % of countries achieving MAPE

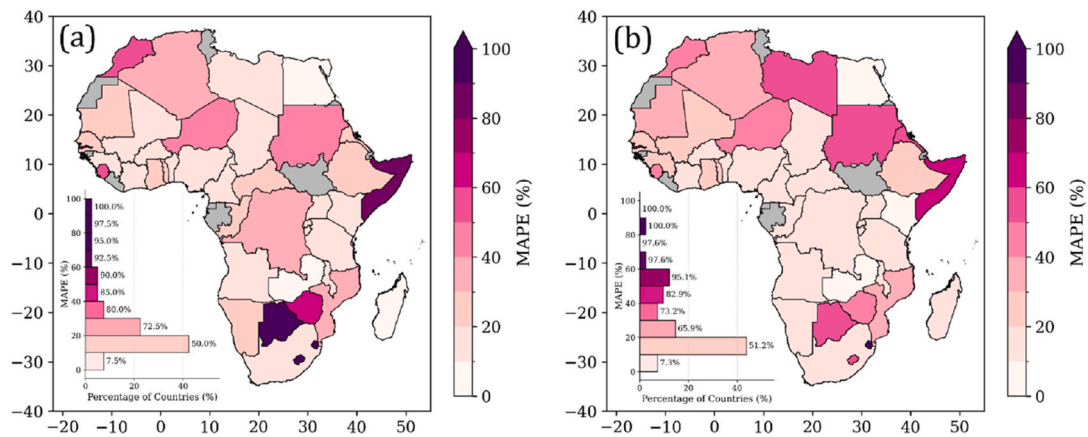


Fig. 9. Spatial distribution of mean absolute percentage error (MAPE) at the national level for maize yield predictions across Africa, comparing the performance of (a) XGBoost and (b) TabPFN models over the 2010–2022 period. MAPE values were computed by aggregating model-predicted yields to the country scale and comparing them with FAOSTAT-reported national yields. Inset bar charts show the proportion of countries falling within specified MAPE intervals. Grey-shaded countries indicate no data values.

values in the 0–30 % range compared to TabPFN. This suggests that XGBoost may be more effective in scaling from local-level predictions to national-level yield estimations.

4.5. Feature contribution and causal effect estimation

To quantify the sensitivity of individual features and establish their causal relationship with yield, we conducted an ablation study and performed causal discovery on the XGBoost model and its predictions, as XGBoost consistently outperformed other models across temporal, local, and national evaluation scales except in Somalia, Zimbabwe, Botswana, and Morocco. The ablation study (Fig. 10) quantifies the sensitivity of different feature groups by measuring the change in predictive performance -represented through ΔR^2 and ΔMAPE -when each feature group is individually removed from the model. The study reveals that the exclusion of remote sensing vegetation indices such as NDVI and NDWI substantially reduces the predictive performance of the model, as R^2 decreased by 0.02 and MAPE increased by 3 % when removing NDVI from the data, and the MAPE value increased by 1 % when removing NDWI. Other features such as soil properties, wind components, minimum temperature, and volumetric soil water also emerged as crucial parameters for accurate yield estimates. As the model was trained using rigorous cross-validation splits, it gives balanced weight to every

individual parameter, avoiding over-reliance on a single feature, which commonly occurs in an overfitting scenario. Therefore, the exclusion of other climatic variables such as precipitation, radiation, and ET_0 leads to minimal or even slightly improved performance, suggesting that their contribution may be either marginal or potentially redundant given the presence of more informative features.

To complement this analysis, we estimated the Average Treatment Effect (ATE) for each input variable using both spatially disaggregated and model-predicted yields (Fig. 11). This comparison allows us to evaluate whether the XGBoost model preserves the meaningful causal relationships present in the data. The estimated ATEs for key features like NDVI, NDWI, and irrigation are strongly positive and consistent across both observed and predicted outcomes, indicating their importance and confirming that the model captures their underlying causal influence. For example, a 0.1 unit increase in NDVI is associated with a yield increase of approximately 0.55 t ha^{-1} in the disaggregated yield data and 0.51 t ha^{-1} in the model's predictions. Negative causal impacts of drought indicators such as SPI-1 and SPI-3 are also accurately captured by the model, further validating its ability to represent real-world agronomic dynamics. Overall, the integration of ablation and causal analyses demonstrates a strong alignment between features that are predictive and those that are causally relevant. This consistency confirms that the XGBoost model not only excels in yield prediction but

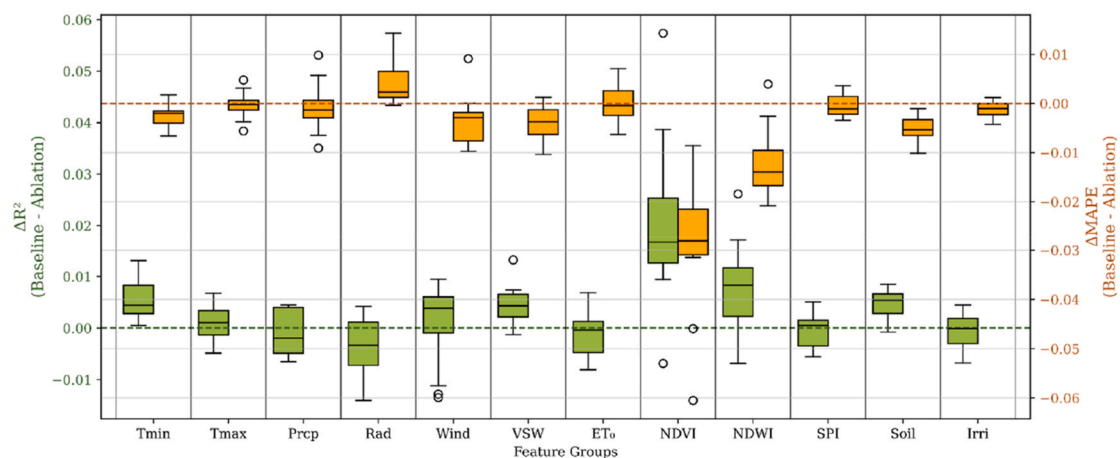


Fig. 10. Impact of feature group ablation on model performance. The figure shows the change in R^2 (ΔR^2) and mean absolute percentage error (ΔMAPE) when individual feature groups are excluded from the XGBoost model. Baseline metrics are derived from the model outputs when trained with the full set of input features, and deviations from this baseline indicate the relative importance of each feature group (refer to Table 1 for feature group description).

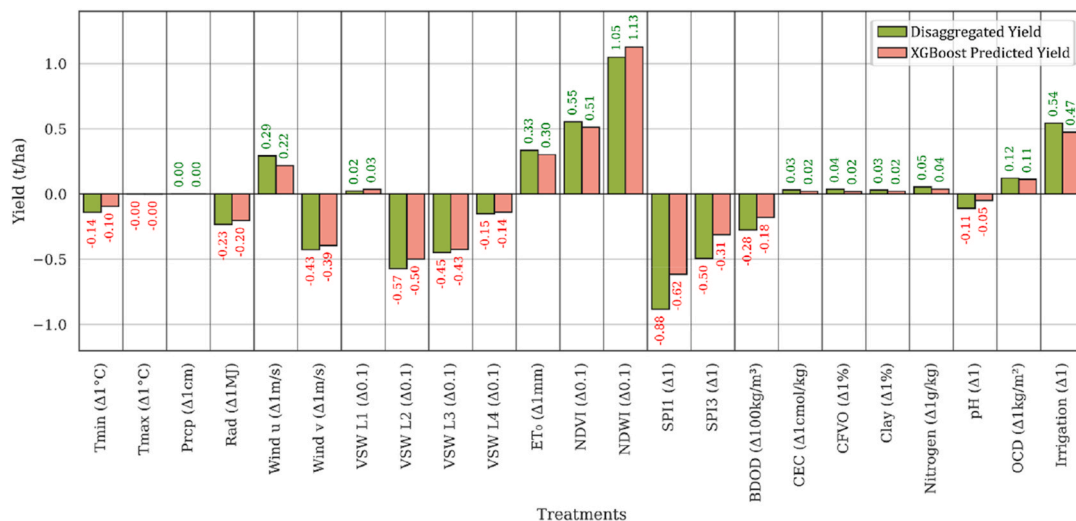


Fig. 11. Average Treatment Effect (ATE) estimates for different input variables. The figure compares the estimated causal effects of individual features on maize yield, using both spatially disaggregated observed yields and XGBoost-predicted yields (refer to Table 1 for feature group description).

also effectively captures the agronomic processes that drive those outcomes.

4.6. Benchmarking against existing gridded yield datasets

To contextualize the performance of the proposed framework and quantify its improvement over existing gridded yield products, we conducted a direct comparison with the widely used Global Dataset of Historical Yields (GDHY). For this assessment, the high-resolution yield estimates from GDHY, XGBoost, and TabPFN were aggregated to the national scale for each year and validated against the official FAOSTAT yield statistics. Rows containing zero yields or missing GDHY values were excluded from the analysis. The results of this benchmarking analysis are presented in Fig. 12.

The comparison reveals a significant performance gap between our models' output and the GDHY dataset. GDHY exhibits a very weak relationship with FAOSTAT yields, with an R^2 of only 0.03, indicating that it captures minimal variance in officially reported national yields across the space and time. Its Mean Absolute Percentage Error (MAPE) is also very high (62.7 %), indicating substantial deviations from the reference data.

In contrast, our data-driven models show a very strong agreement with FAOSTAT yields. The XGBoost model achieves an R^2 of 0.88 with a MAPE of 27.4 %, while the TabPFN model performs slightly better in this case, reaching an R^2 of 0.89 and a MAPE of 25.7 %. These results highlight the substantial accuracy gains achieved by the proposed

framework relative to existing global gridded yield products.

5. Discussion

5.1. Significance and adaptability of high-resolution maize yield prediction framework

The ability to generate continent-scale, high-resolution crop yield prediction maps marks a major advancement in agricultural monitoring, especially for regions like Africa where field-level data are scarce and yield heterogeneity is high. In this study, we present a novel framework that combines synthetic yield disaggregation with machine learning and Earth Observation (EO) data, offering an innovative solution to the chronic spatial data gaps in African agriculture. Unlike traditional assessments based on coarse national-level statistics, our method captures the fine-scale spatial and temporal variability critical for identifying yield gaps, evaluating food security risks, and assessing localized climate impacts (Aramburu-Merlos et al., 2024). Our approach generates pixel-level maize yield estimates using a synthetic disaggregation technique driven by Net Primary Productivity (NPP) and multi-source EO data. The resulting high-resolution maize yield maps (Fig. 13) demonstrate strong consistency with officially reported national statistics, validating the method's effectiveness. By simulating disaggregated training targets, we enable the use of advanced models such as XGBoost and TabPFN, which show strong performance across diverse agro-climatic zones. Notably, our analysis reveals significant regional

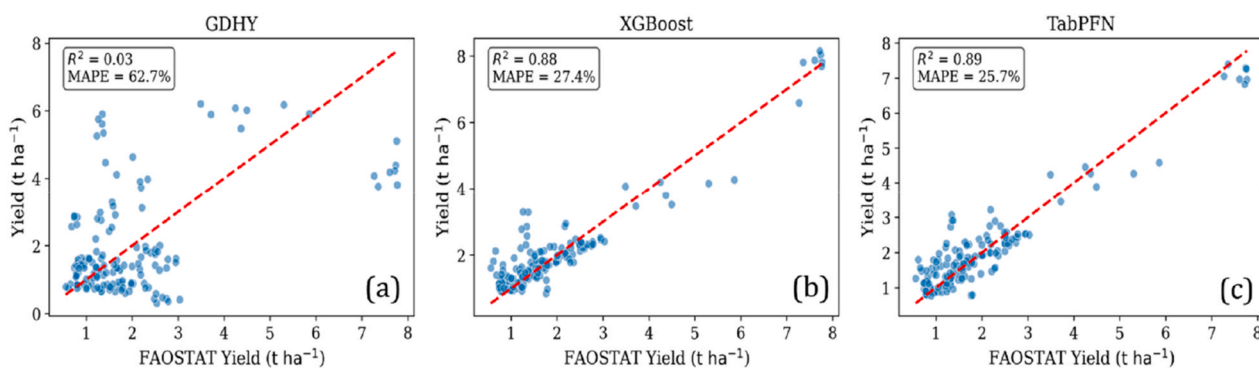


Fig. 12. Validation of nationally aggregated yields from (a) GDHY, (b) XGBoost, and (c) TabPFN against FAOSTAT data. Both XGBoost ($R^2 = 0.88$) and TabPFN ($R^2 = 0.89$) show a vastly superior correlation and lower error compared to the GDHY baseline ($R^2 = 0.03$).

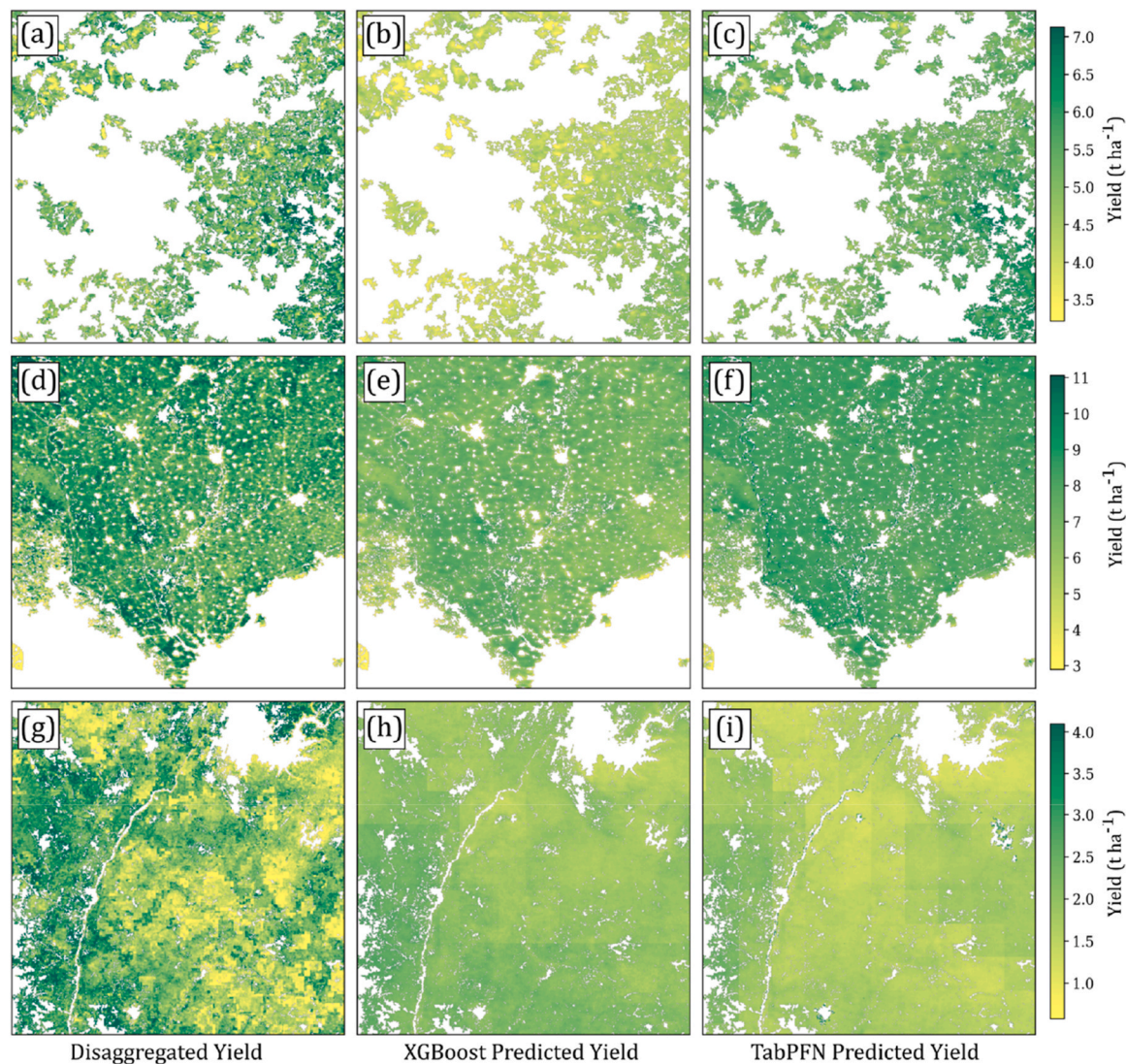


Fig. 13. Visual comparison of predicted maize yield and disaggregated reference yield across three representative regions in Africa for the year 2022. The first row (a–c) corresponds to a region in South Africa (medium yield), the second row (d–f) to a region in Egypt (high yield), and the third row (g–i) to a region in Nigeria (low yield). Columns show (from left to right): disaggregated reference yield, predicted yield from XGBoost, and predicted yield from TabPFN. All maps were generated at a 250 m spatial resolution (with a total area in each map $\sim 1\text{deg} \times 1\text{deg}$), matching the resolution of the WAPOR Net Primary Production (NPP) data and the vegetation indices (e.g., NDVI and NDWI) available for training. Yield values are expressed in tonnes per hectare (t ha^{-1}). The models demonstrate strong agreement with reference data in high and medium-yielding regions such as Egypt and South Africa (b–f), whereas prediction accuracy declines in low-yielding regions like parts of Nigeria (g–i). These patterns underscore the models' robustness in agriculturally intensive areas with more stable yield dynamics and less climatic variability.

differences in model generalization, reinforcing the importance of spatially explicit model evaluation beyond administrative boundaries. Crucially, the model training pipeline was built around crop development phase features and implemented using a rigorous expanding window cross-validation strategy. This approach avoids common pitfalls associated with random or stratified splits that can introduce temporal leakage and artificially inflate performance. By preserving the chronological structure of training and testing data, we simulate realistic prediction scenarios that better reflect operational decision-making conditions under evolving climatic and environmental pressures. To our knowledge, this is the first study to produce high-resolution yield predictions at a pan-African scale using a fully synthetic yield framework. Most prior efforts have focused on single countries or regions and relied on labor-intensive field yield surveys (Leroux et al., 2019; Adisa et al., 2019; Asamoah et al., 2024). In contrast, we combine scalable synthetic data generation with a temporally consistent evaluation protocol, retraining models in each fold to prevent overfitting. Our results meet or exceed the accuracies reported in previous studies. For instance (Leroux

et al., 2019), reported R^2 values of 0.42–0.53 in West Africa (Adisa et al., 2019); reported 0.81 in Ghana; and (Asamoah et al., 2024) achieved 0.67–0.86 in Southern Africa for 2018–2019. However, because these studies use different datasets and timeframes, direct year-to-year comparisons are not meaningful.

Instead, our framework provides a more generalizable performance summary across years and countries. As shown in Fig. 8, our models explain 78–79 % of yield variance continent-wide, with annual performance ranging from 50 % to over 90 %, depending on climatic conditions and data density. These findings underscore the robustness, scalability, and policy relevance of our yield mapping framework in supporting data-driven agricultural planning, early warning systems, and sustainable intensification efforts across Africa. Performance nevertheless degrades in anomalous seasons, indicating that future work should incorporate hybrid approaches that blend process-based crop models with machine- and deep-learning techniques to better capture extremes.

In this light, the integration of high-resolution synthetic yield data

with temporally aware validation strategies represents a significant advancement in data-driven precision agriculture. It narrows the gap between broad-scale statistical reporting and localized decision support, thereby advancing more resilient and adaptive agricultural systems across Africa.

5.2. Importance of vegetation indices and climate signals in yield prediction

Our findings underscore the critical role of remote sensing variables—particularly vegetation indices—in improving the accuracy, interpretability, and robustness of crop yield predictions across heterogeneous African landscapes. Among these, NDVI and NDWI emerged as the most influential predictors, as demonstrated through both ablation studies and causal effect estimation. When these indices were removed, predictive performance declined noticeably, confirming their essential contribution. These results align with earlier research: for instance (Xu and Katchova, 2019), showed that NDVI captures nonlinear relationships with soybean yield across phenological stages, while (Tamás et al., 2023; Teal et al., 2006) demonstrated its effectiveness in predicting corn grain yield. Our work builds on these insights by not only validating their utility at continental scale but also quantifying their causal impact on yield—thus advancing understanding of how vegetation health and moisture availability mediate crop outcomes under variable climate conditions.

In addition to vegetation indices, this study employed Net Primary Productivity (NPP) as both a disaggregation proxy and an ecologically meaningful indicator of productivity. This dual role is central to our approach: NPP guided the spatial disaggregation of national yield statistics into high-resolution synthetic targets and provided a biophysically relevant feature for yield modelling. This enabled machine learning models to learn from both macro-level productivity patterns and fine-scale environmental variability, enhancing predictive power across diverse agro-ecological contexts. Our causal analysis further revealed that climate extremes—particularly drought indicators such as SPI-1 and SPI-3 have strong negative effects on yield. This insight is consistent with the observed decline in model accuracy between 2019 and 2021, which coincided with widespread rainfall deficits across sub-Saharan Africa (Ayugi et al., 2022).

These events exposed vulnerabilities in rainfed agricultural systems and demonstrated the urgent need for climate-resilient strategies, including early warning systems, drought-tolerant crops, and adaptive land management.

Taken together, these findings highlight the novel integration of causal inference, remote sensing, and climate extremes in our framework. By embedding such variables into yield prediction models, we enable more accurate prediction, enhance explanatory depth, and provide actionable insights for resilience planning under increasing climatic uncertainty.

5.3. Potential of tabular foundation models for crop yield prediction

The advent of Tabular Foundation Models (TFMs), particularly the TabPFN model, represents a paradigm shift in data-driven approaches to crop yield prediction. In contrast to traditional machine learning and deep learning models that necessitate extensive dataset-specific training, TabPFN employs in-context learning and is pretrained on millions of synthetically generated datasets derived from structural causal models (SCMs). This pretraining strategy allows TabPFN to generalize effectively across diverse tabular prediction tasks, requiring little to no task-specific fine-tuning.

Our results indicate that TabPFN delivers competitive performance even when trained on substantially smaller datasets compared to other models such as XGBoost, LightGBM, and HDNN. A key challenge in crop yield prediction, particularly in developing countries such as those in Africa, is the limited availability of large, high-quality training datasets.

The absence of openly accessible and large-scale datasets suitable for deep learning has significantly constrained the advancement of accurate predictive models for crop yield (Lin et al., 2024). In this context, TabPFN offers a distinct advantage: unlike conventional deep learning approaches, it does not rely on large training datasets. Despite its minimal data requirements, TabPFN consistently demonstrated strong spatial prediction accuracy (Figs. 7 and 13), especially in regions characterized by data scarcity and agro-climatic heterogeneity. Notably, it outperformed traditional models in reducing mean absolute percentage error (MAPE) and more effectively captured fine-scale yield variability across the African continent with higher spatial resolution.

TabPFN also has the unique advantage of producing probabilistic predictions, which sets it apart from most traditional models. This means it can estimate the uncertainty in its predictions, which is critical for agricultural decision-making under climatic volatility. Since crop yields are influenced by many unpredictable factors, having a model that reflects this uncertainty can improve risk assessments and support better early warning systems. That said, Tabular Foundation Models like TabPFN are not without limitations. The current version used in our study can handle only up to 10,000 rows and 500 columns, which can be a constraint when working with larger or more complex datasets. To stay within this limit, we had to sample 10,000 data points from each training fold. As the amount of data and the number of years increased, it became harder to ensure that the sample fully represented the original dataset. This has likely contributed to TabPFN's drop in performance in recent years, where models like XGBoost performed better. Additionally, because TabPFN is based on a transformer architecture and uses a full forward pass for each prediction, it can be quite demanding in terms of GPU memory and computational resources—especially if applied at scale or in real-time settings.

Despite their current limitations, Tabular Foundation Models (TFMs) represent a promising direction for crop yield prediction. Their ability to generalize well across diverse regions, perform effectively with limited training data, and provide built-in uncertainty estimates makes them especially valuable in data-scarce and agro-ecologically diverse settings. As future releases of these models overcome constraints related to input size and feature dimensionality, their potential for large-scale, high-resolution agricultural monitoring will likely grow.

5.4. Limitation and uncertainty of the study

A fundamental aspect of our study involves disaggregating national-level FAO maize yield statistics into high-resolution, pixel-level yield estimates by employing Net Primary Productivity (NPP) as a proxy. This strategy enables the development of spatially explicit models at scale in contexts where field-level observations are lacking, but it also introduces critical uncertainties that must be considered when interpreting the results. Notably, the availability of NPP data is restricted both in time and space. Accessing long-term NPP time series data across Africa—or globally—is particularly challenging due to limited open-source data availability. Consequently, our analysis was confined to the 2009–2022 period, corresponding to the temporal coverage of the WAPOR dekadal NPP dataset, which is available exclusively for Africa. This restriction precluded the use of earlier FAO yield records that could have extended the training period and enhanced the model's generalizability. Additionally, the assumption that NPP exhibits a consistent and linear relationship with maize yield across diverse agroecological zones may not always be valid. Crop yields are influenced by a variety of factors beyond biomass production, such as nutrient limitations, pest infestations, and stress conditions occurring after anthesis. This implies that in complex, heterogeneous farming systems, our yield estimates should be interpreted as a robust indicator of vegetative productivity rather than a precise prediction of farm-level grain yield and should ideally be used in conjunction with local agronomic knowledge.

Secondly, while our model integrates a diverse set of Earth Observation (EO) features—including climate, vegetation, and soil parameters—

some of these inputs are derived from coarse-resolution datasets. For example, climate data from ERA5-Land are available at an approximate spatial resolution of 9 km, which is considerably coarser than the 250 m resolution of the final yield estimates. This disparity in spatial scales can introduce artefacts in the model outputs, particularly in regions characterized by sharp climatic gradients. Consequently, while the maps are highly valuable for regional assessment and identifying broad patterns, their application for field-scale precision agriculture or in areas with complex topography should be approached with caution, as localized microclimate may not be fully resolved. Such effects are especially pronounced in low-yielding areas of Nigeria, as illustrated in Fig. 13h and i. The mismatch in resolution may lead to spatial smoothing or misrepresentation of fine-scale microclimatic conditions that are critical to crop growth and yield formation.

Third, our spatial evaluation results (Fig. 7) indicate that model prediction errors are substantially higher in climatically unstable regions such as the Sahel, the Horn of Africa, and in small countries with complex terrain. These regions experience highly erratic rainfall, frequent droughts, and diverse agroecological conditions, often combined with sparse and fragmented agricultural land. Such environmental variability limits the model's generalizability, especially when it has been trained on data from more stable agro-climatic zones such as South Africa and Egypt. In addition, the lack of accurate, high-resolution cropland maps—along with persistent misclassification of agricultural areas—further introduces noise in the data. This is particularly evident in countries like Lesotho, Eswatini, Botswana, Somalia, Sudan, Niger, Eritrea, and Libya, where these challenges contribute to higher prediction MAPE and diminished accuracy. Consequently, a key limitation of our framework is that its performance and findings are not directly generalizable to the excluded, data-poorest nations. While the model is robust across the vast majority of Africa's agricultural land, its applicability in these specific data-scarce regions remains an important area for future investigation.

Fourth, Spatial yield variability may need a much higher resolution than 250 m due to the small farm and field sizes (smallholder farming) in several parts of Africa and crops are also grown in mixtures, e.g. agroforestry, in some places.

Finally, our model does not account for agricultural management practices—such as fertilizer application, sowing techniques, and pest control—or socio-economic factors like market accessibility and land tenure arrangements, all of which are known to play a critical role in shaping yield outcomes. The lack of harmonized, spatially comprehensive datasets capturing these dimensions across Africa presents a key limitation, preventing a more holistic representation of the factors driving yield variability. As a result, the model may overlook important context-specific influences, particularly within smallholder farming systems where such variables are especially consequential.

5.5. Practical implications for policy and planning

Beyond its methodological advancements, the proposed high-resolution yield mapping framework carries substantial practical relevance for strengthening food security and agricultural resilience across Africa. For national policymakers, the generated yield maps provide a strong evidence base for designing targeted interventions, facilitating the strategic allocation of resources such as subsidies and irrigation investments toward identified productivity hotspots, and informing the management of national grain reserves. At the subnational level, agricultural planners and extension services can exploit the 250 m spatial detail to pinpoint communities most vulnerable to climatic shocks, guide the dissemination of climate-resilient crop varieties, and deliver site-specific agronomic recommendations. Furthermore, the framework serves as a powerful early warning tool for humanitarian organizations, enabling them to anticipate food deficits and deploy aid more efficiently. The data also holds commercial value for the private sector, including the development of more accurate index-based insurance schemes for

smallholder farmers and improved risk assessment models for agricultural credit provision. By translating complex Earth Observation data into actionable, high-resolution yield intelligence, our framework helps bridge the gap between scientific research and on-the-ground decision-making, supporting a more data-driven approach to agricultural development in Africa.

6. Conclusion

This study presents a scalable, data-driven framework for high-resolution maize yield prediction across 42 African countries by integrating multi-source Earth Observation (EO) data, synthetic yield disaggregation based on Net Primary Productivity (NPP), and advanced machine learning techniques. By aligning input features with crop phenology and employing an expanding window cross-validation strategy, our framework generates pixel-level yield estimates that capture spatial heterogeneity often missing in administrative-level statistics.

Among the models evaluated, XGBoost demonstrated strong and consistent temporal performance, while TabPFN excelled in spatial generalization, particularly in climatically diverse and data-sparse regions. The incorporation of EO-derived variables—such as NDVI, NDWI, evapotranspiration, and drought indices—together with irrigation practices and some climate data, significantly enhanced predictive accuracy. Further, ablation and causal inference analyses helped us better understand how different groups of features contribute to the model accuracy. Validation against national yield data from FAOSTAT confirmed the alignment of our high-resolution predictions with officially reported trends. Nonetheless, several limitations remain. The use of NPP for yield disaggregation, while effective, assumes stable biomass–yield relationships that may vary across agroecological zones. Temporal constraints in NPP availability restricted our modelling to 2009–2022, limiting historical comparisons. The coarse spatial resolution of climate inputs (e.g., ERA5-Land) also introduces spatial mismatches that can reduce pixel-level accuracy. Additionally, model performance declined in climatically volatile regions such as the Sahel, where sparse cropland distribution and high interannual variability challenge model generalizability. Architectural limitations of TabPFN, including fixed training sample sizes and high computational costs, further constrained its application in more recent years. Looking ahead, future research should prioritize three key directions to improve the framework's accuracy, reliability, and applicability. First, integrating ground-based observations is essential to validate the NPP-based yield disaggregation approach, reducing dependence on empirical biomass–yield relationships. Incorporating detailed agricultural management variables—such as fertilizer application rates, cultivar information, and precise planting dates—alongside socio-economic factors like market access, will help capture finer-scale drivers of yield variability. Third, improving the spatial resolution of climate data is critical to minimizing inconsistencies with Earth Observation datasets. This can be achieved through statistical downscaling of coarse products (e.g., ERA5-Land), advanced super-resolution deep learning approaches, and by leveraging new high-resolution satellite-derived climate datasets to enhance pixel-level accuracy, particularly in environmentally diverse regions. As Tabular Foundation Models continue to evolve, upcoming versions with higher capacity and scalability may overcome current input limitations, unlocking their full potential for continent-wide monitoring. Incorporating forecasted climate scenarios will be essential to shift from retrospective analysis toward real-time prediction and future yield projections, thereby enhancing early warning systems and policy responsiveness.

In conclusion, this study demonstrates the feasibility, novelty, and practical value of a high-resolution, machine learning-enabled prediction framework tailored to Africa's data-scarce yet climatically dynamic agricultural landscapes. With continued methodological innovation and deeper data integration, such approaches offer transformative potential

for supporting climate-resilient agricultural planning and improving food security across the continent.

CRediT authorship contribution statement

Krishnagopal Halder: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Frank Ewert:** Writing – review & editing. **Anitabha Ghosh:** Writing – original draft. **Kaushik Muduchuru:** Writing – review & editing, Writing – original draft. **Lily-belle Sweet:** Writing – review & editing. **Radwa Elshaw:** Writing – review & editing. **Jan Timko:** Writing – review & editing. **Wenhi Zheng:** Writing – review & editing. **Karam Alsafadi:** Writing – review & editing. **Gang Zhao:** Writing – review & editing. **Michael Maerker:** Writing – review & editing. **Manmeet Singh:** Writing – review & editing. **Lei Guoging:** Writing – review & editing. **Thomas Gaiser:** Writing – review & editing. **Dominik Behrend:** Writing – review & editing. **Yue Shi:** Writing – review & editing. **Liangxiu Han:** Writing – review & editing. **Masahiro Ryo:** Writing – review & editing. **Amit Kumar Srivastava:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Code availability

All Python notebooks developed in this study are publicly available at the following GitHub repository: <https://github.com/geonextgis/High-Resolution-Maize-Yield-Mapping-Africa>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.srs.2025.100344>.

Data availability

Data will be made available on request.

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